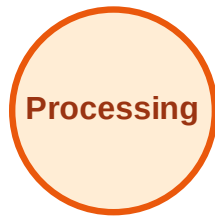


BFS Traversal

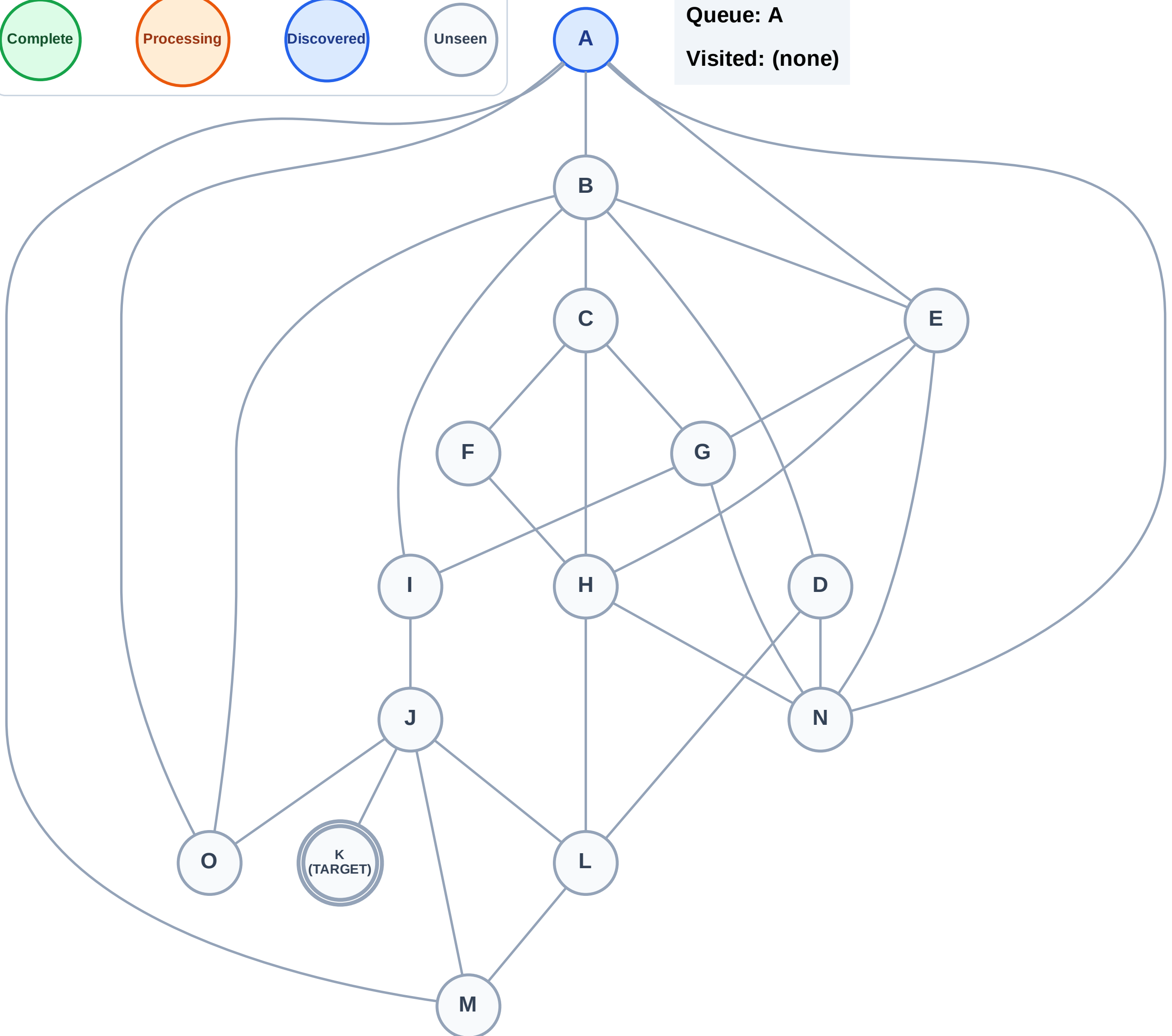
Step 1: Start at A (Seeking target node K (Eager Short-Circuit))

Legend



Queue: A

Visited: (none)



BFS Traversal

Step 2: Process node A

Legend

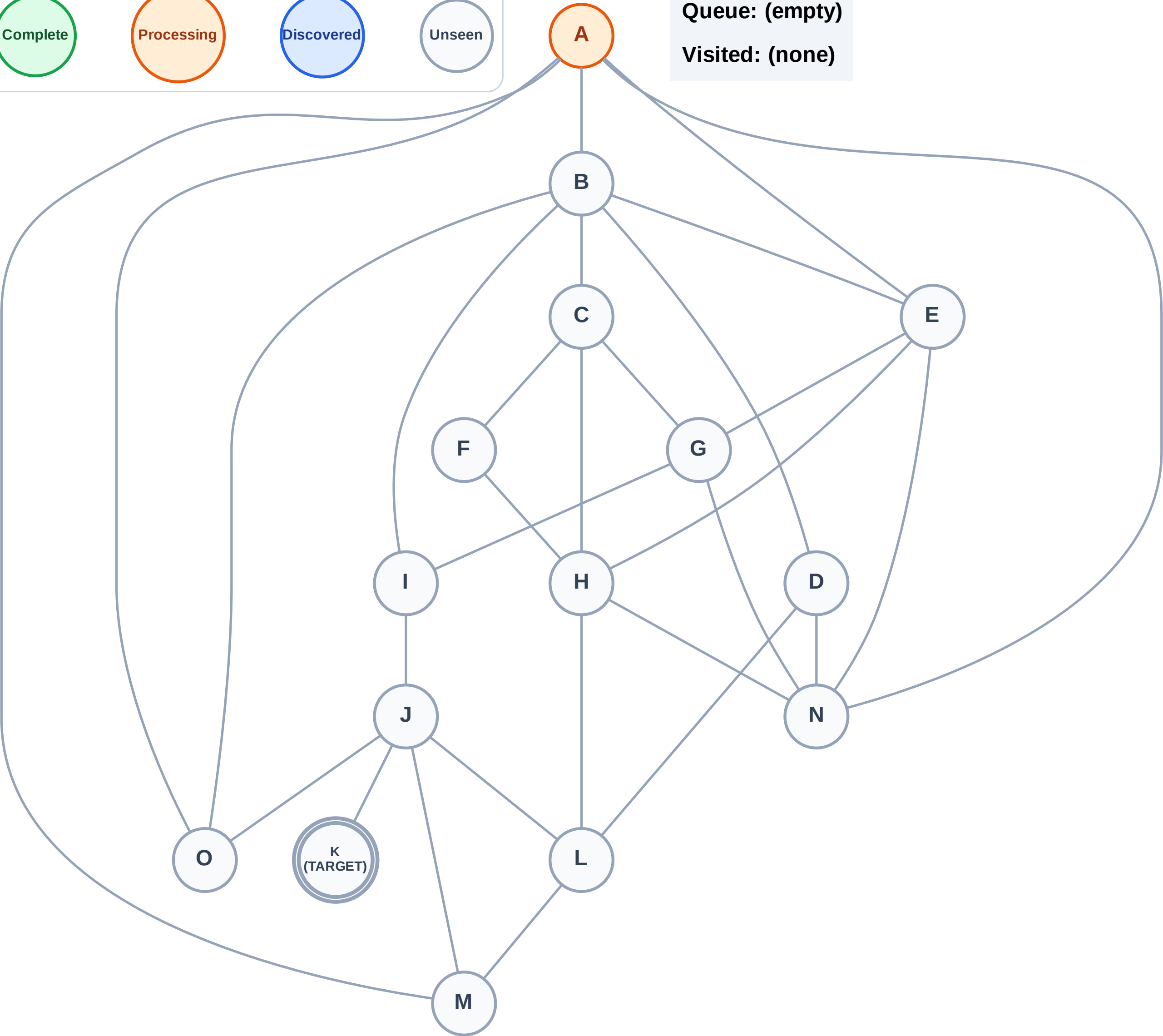
Complete

Processing

Discovered

Unseen

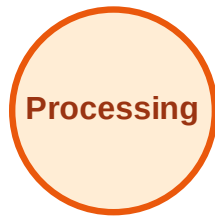
Queue: (empty)
Visited: (none)



BFS Traversal

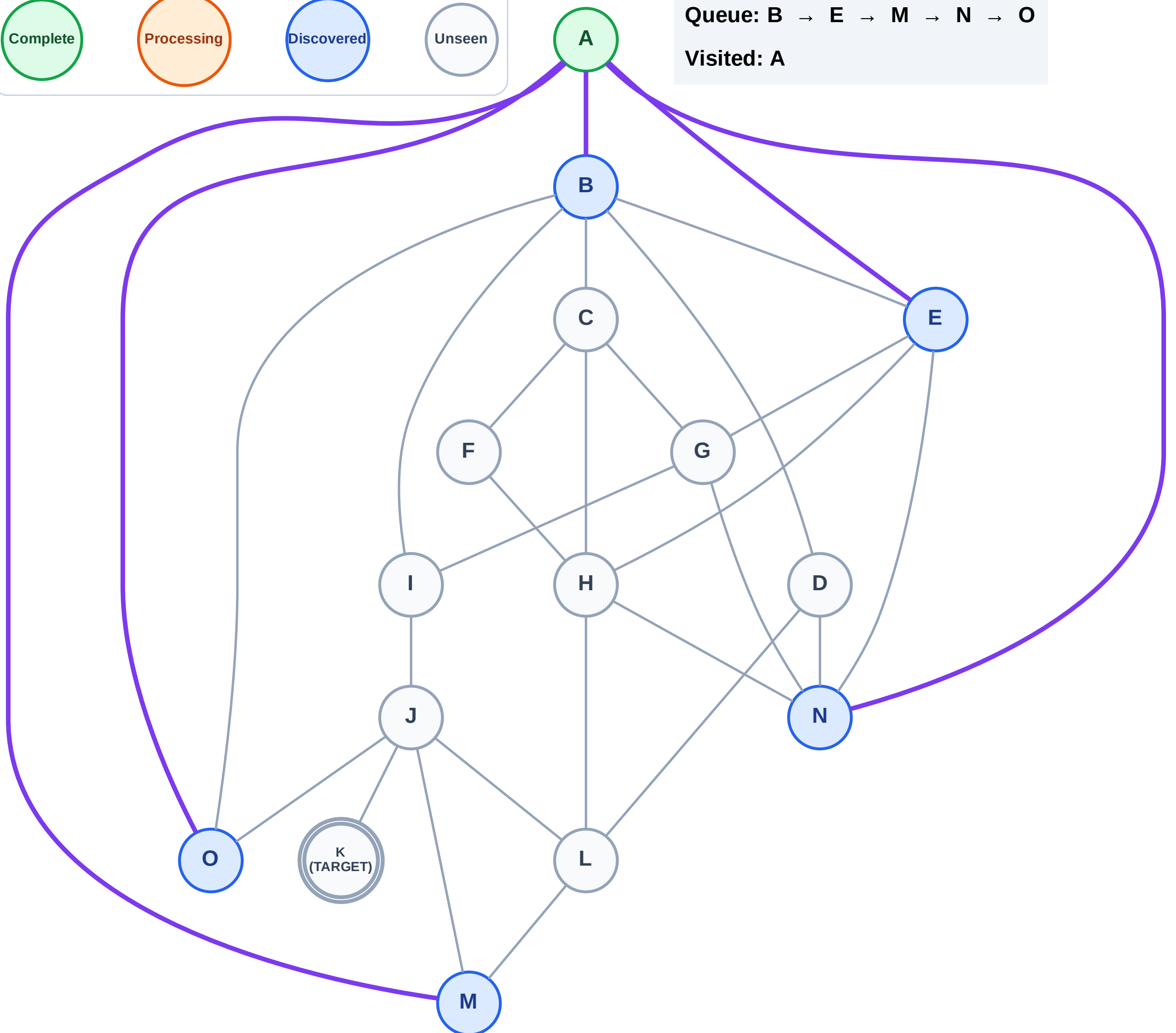
Step 3: Discover B, E, M, N, O from A

Legend



Queue: B → E → M → N → O

Visited: A



BFS Traversal

Step 4: Process node B

Legend

Complete

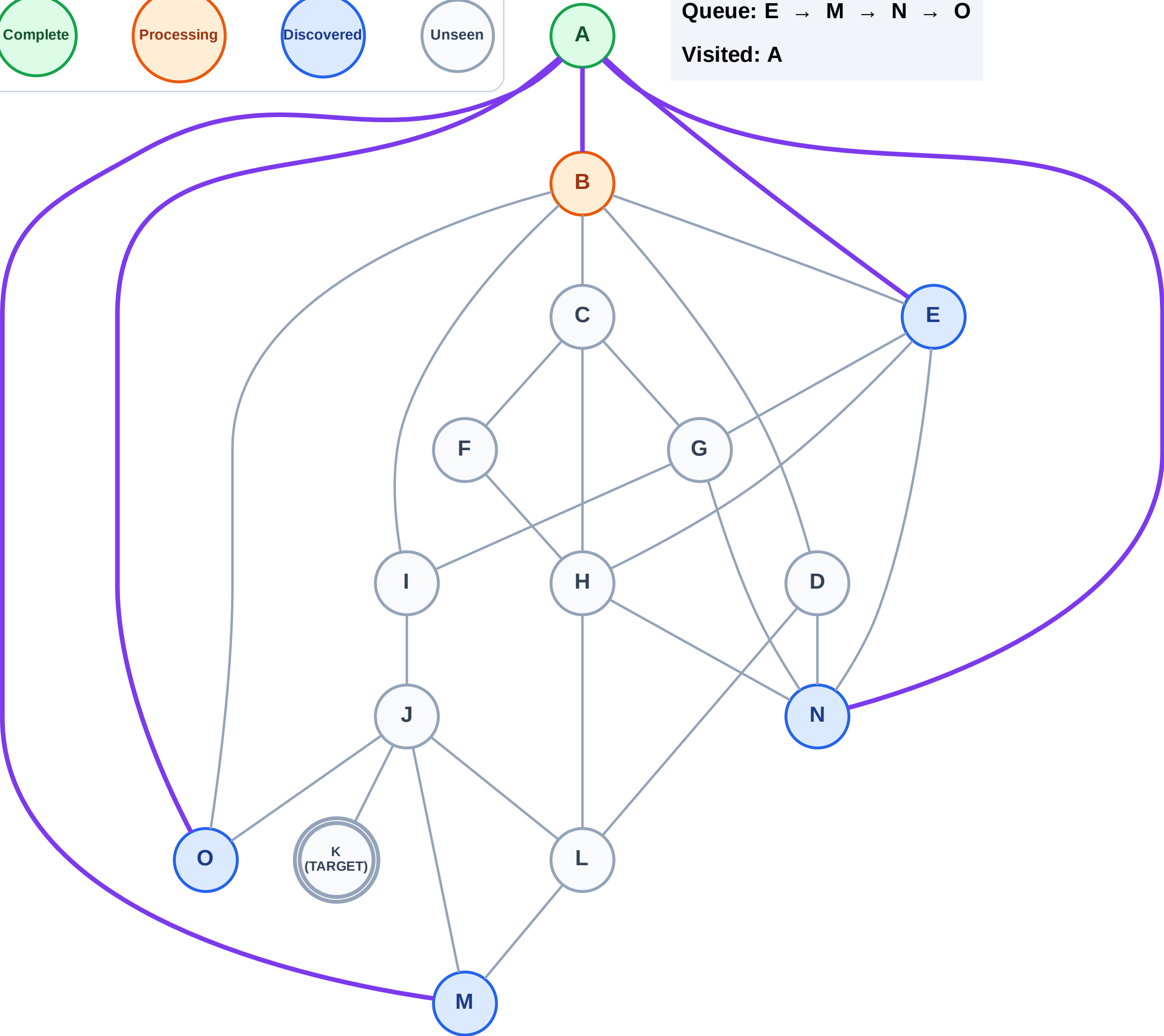
Processing

Discovered

Unseen

Queue: E → M → N → O

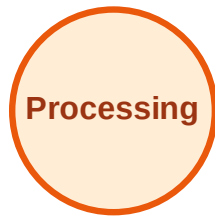
Visited: A



BFS Traversal

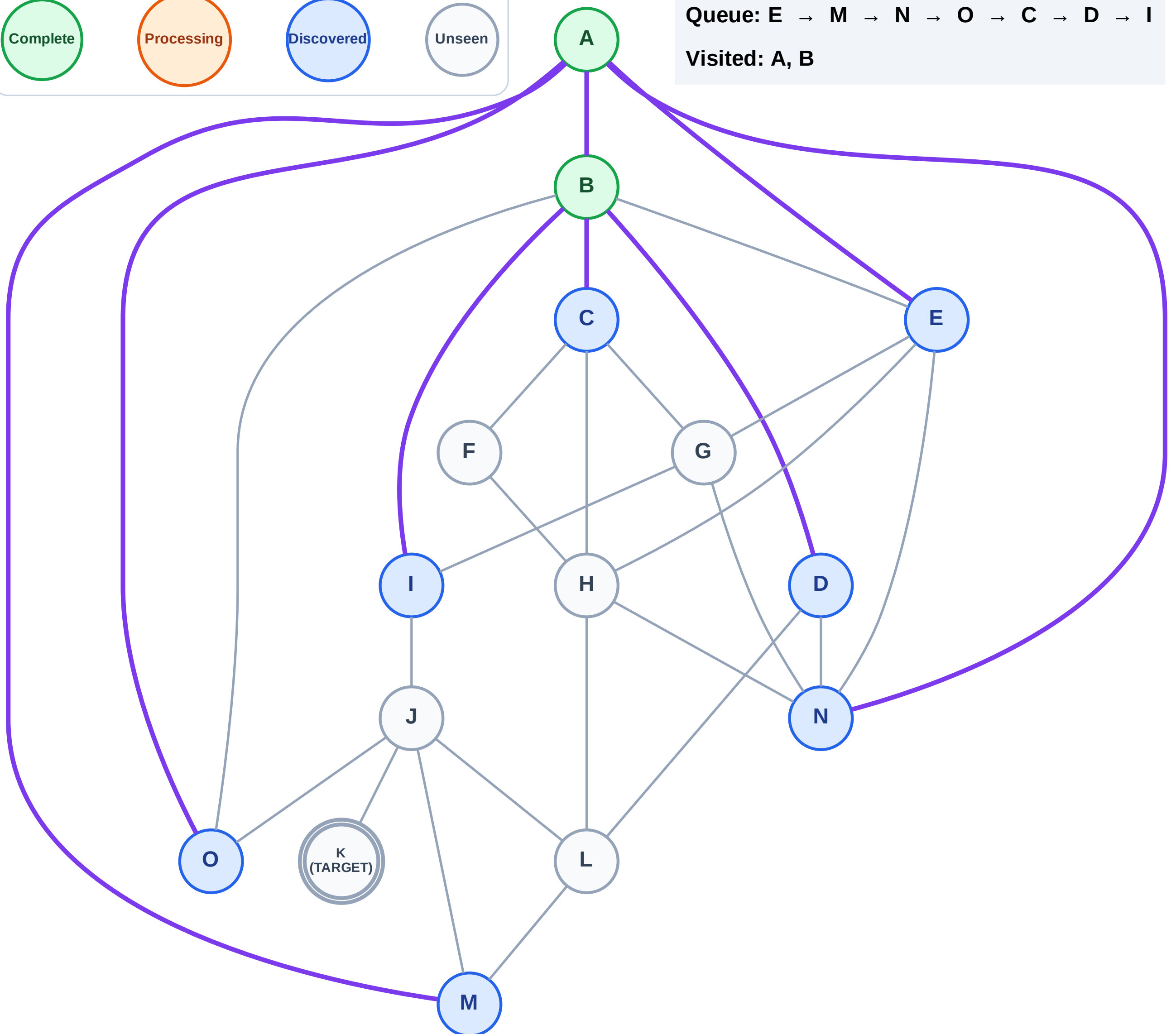
Step 5: Discover C, D, I from B

Legend



Queue: E → M → N → O → C → D → I

Visited: A, B



BFS Traversal

Step 6: Process node E

Legend

Complete

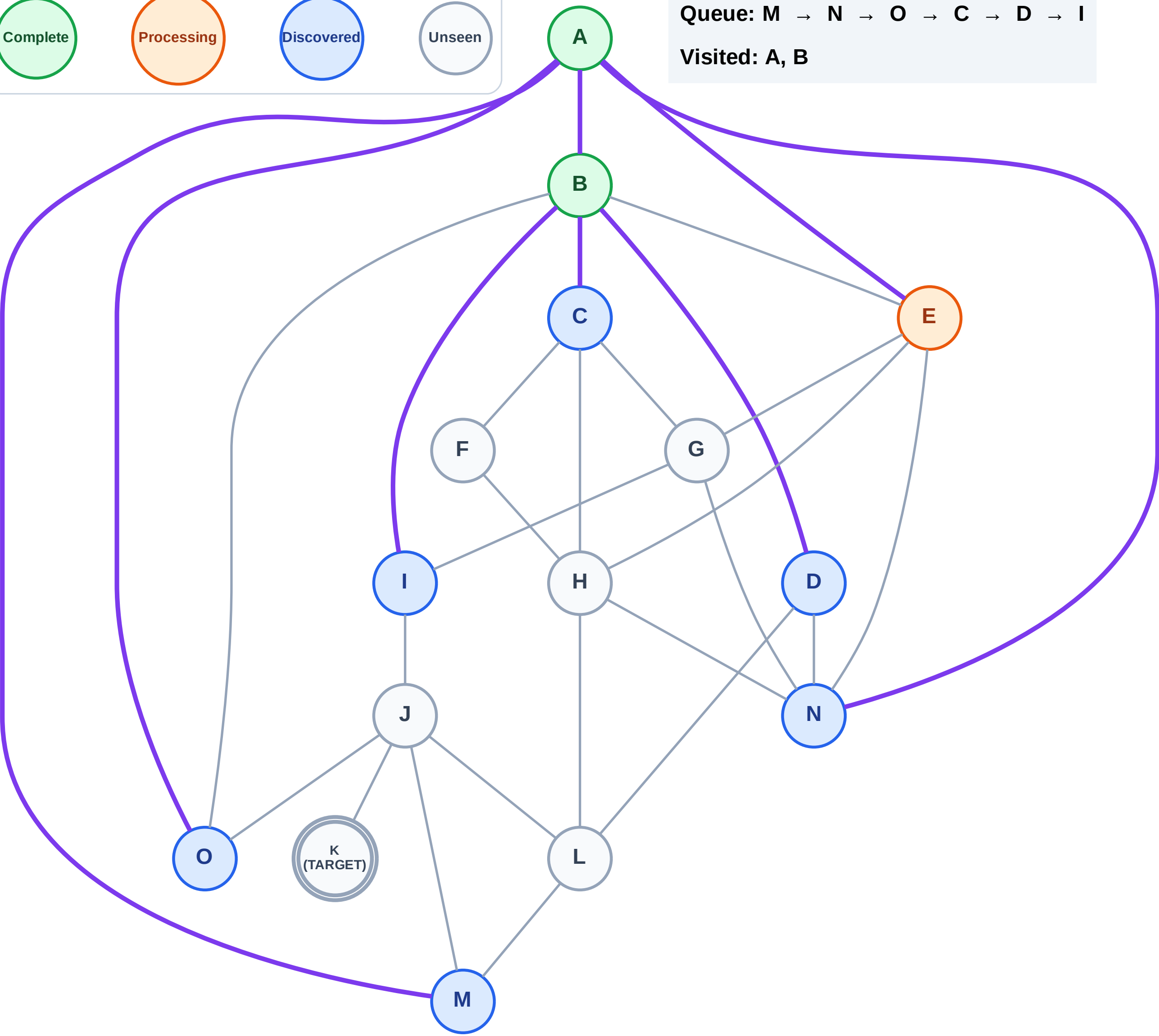
Processing

Discovered

Unseen

Queue: M → N → O → C → D → I

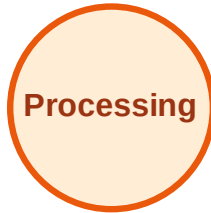
Visited: A, B



BFS Traversal

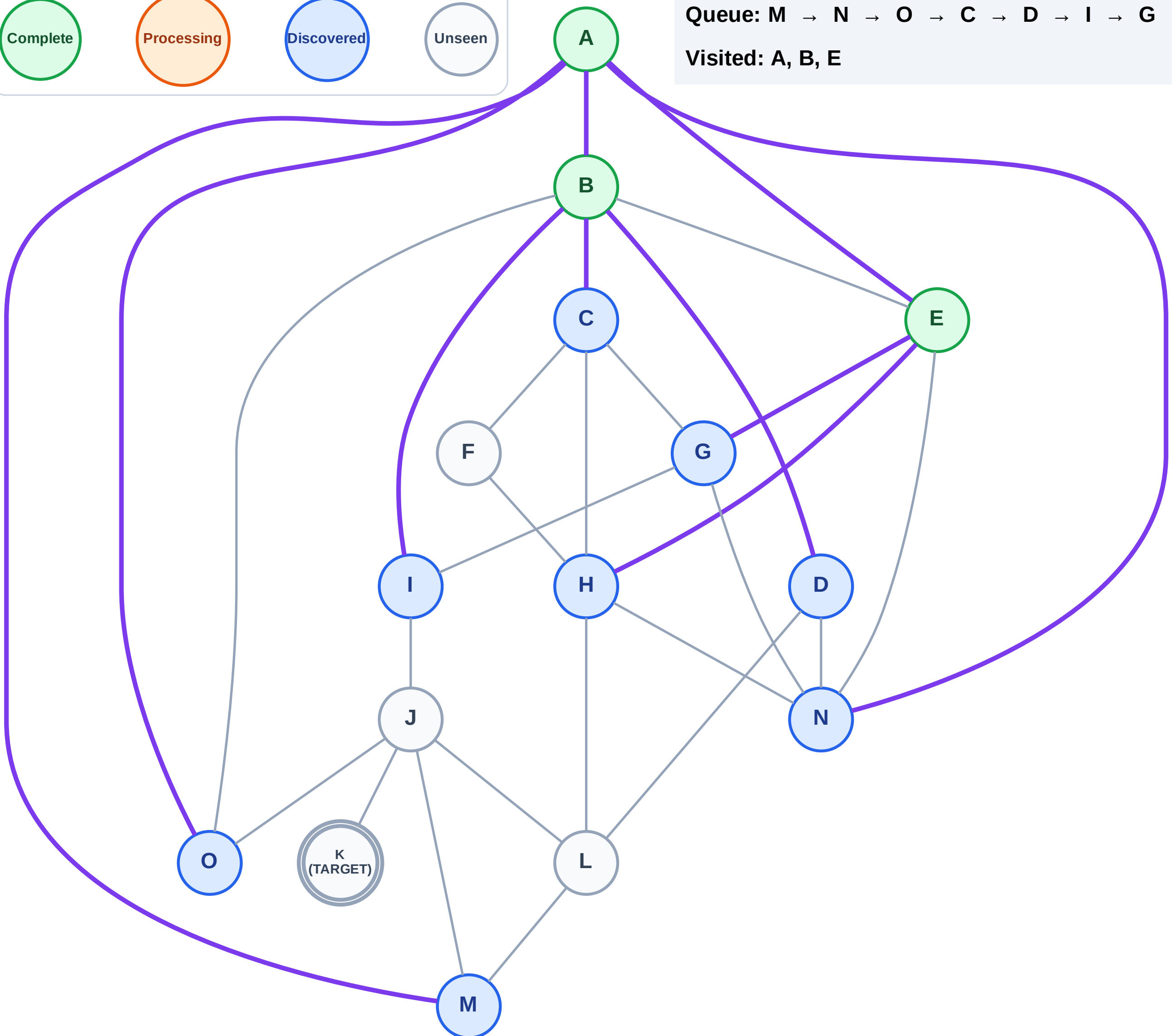
Step 7: Discover G, H from E

Legend



Queue: M → N → O → C → D → I → G → H

Visited: A, B, E



BFS Traversal

Step 8: Process node M

Legend

Complete

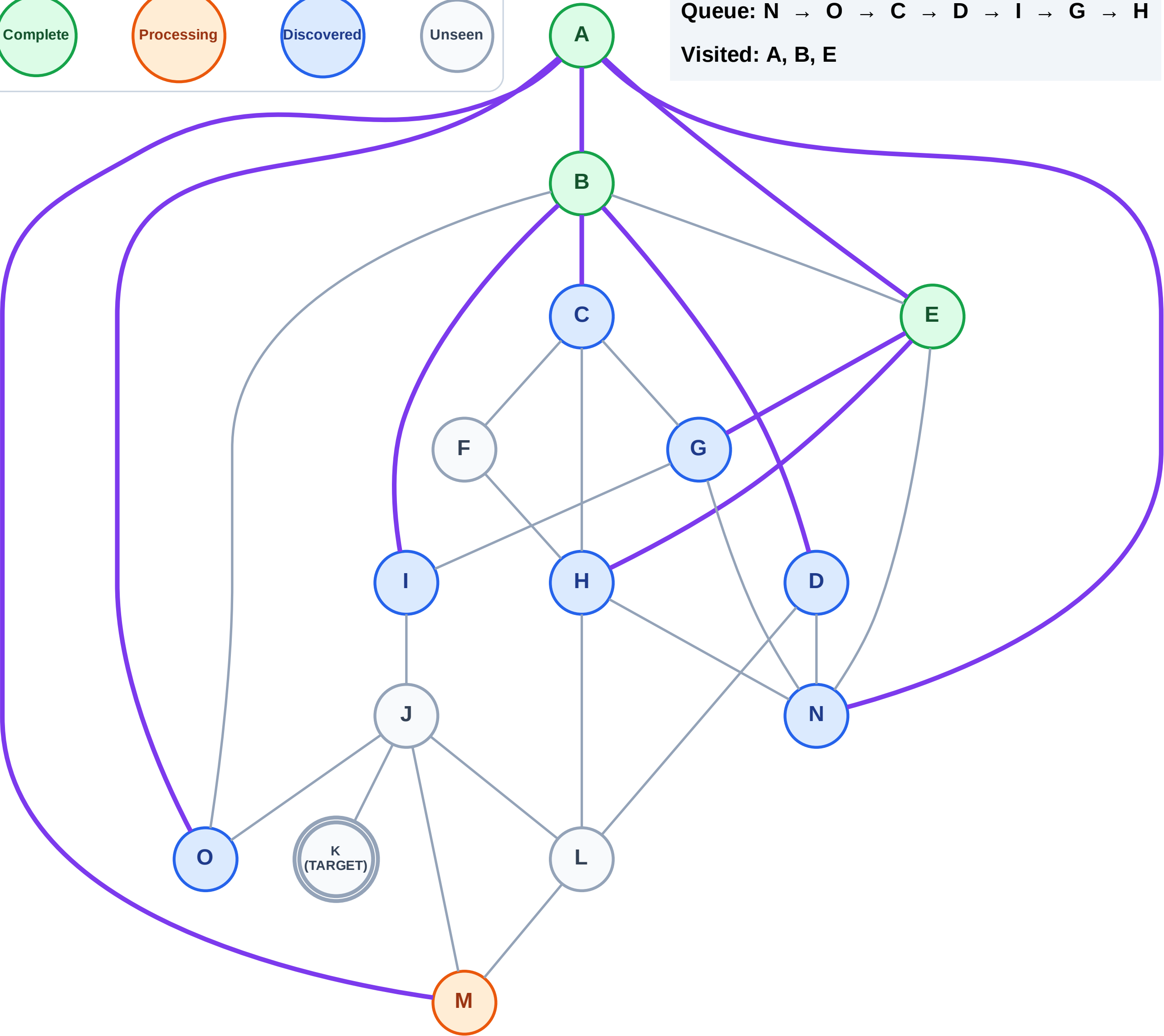
Processing

Discovered

Unseen

Queue: N → O → C → D → I → G → H

Visited: A, B, E



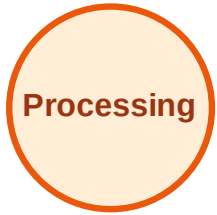
BFS Traversal

Step 9: Discover J, L from M

Legend



Complete



Processing



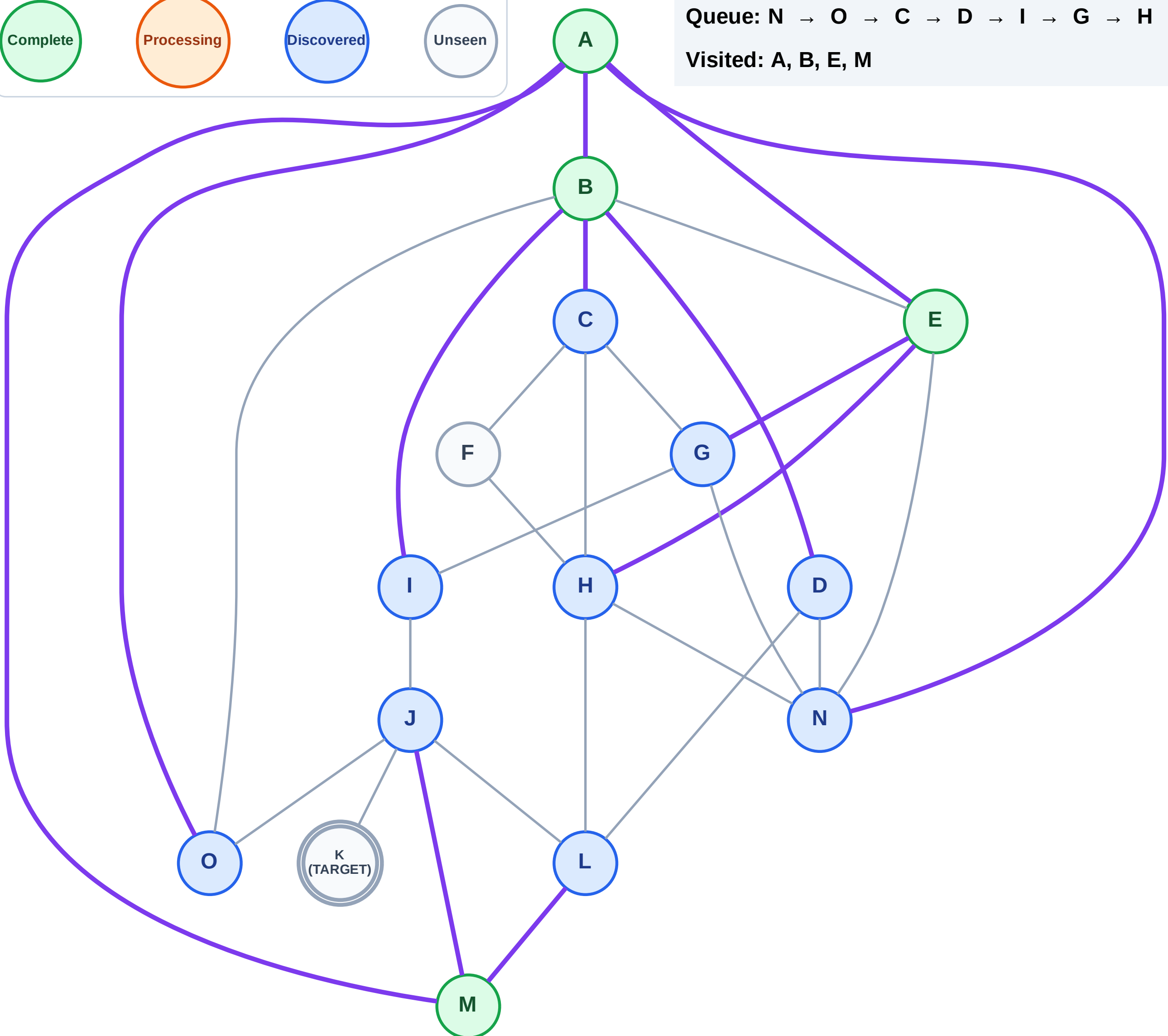
Discovered



Unseen

Queue: N → O → C → D → I → G → H → J → L

Visited: A, B, E, M



BFS Traversal

Step 10: Process node N

Legend

Complete

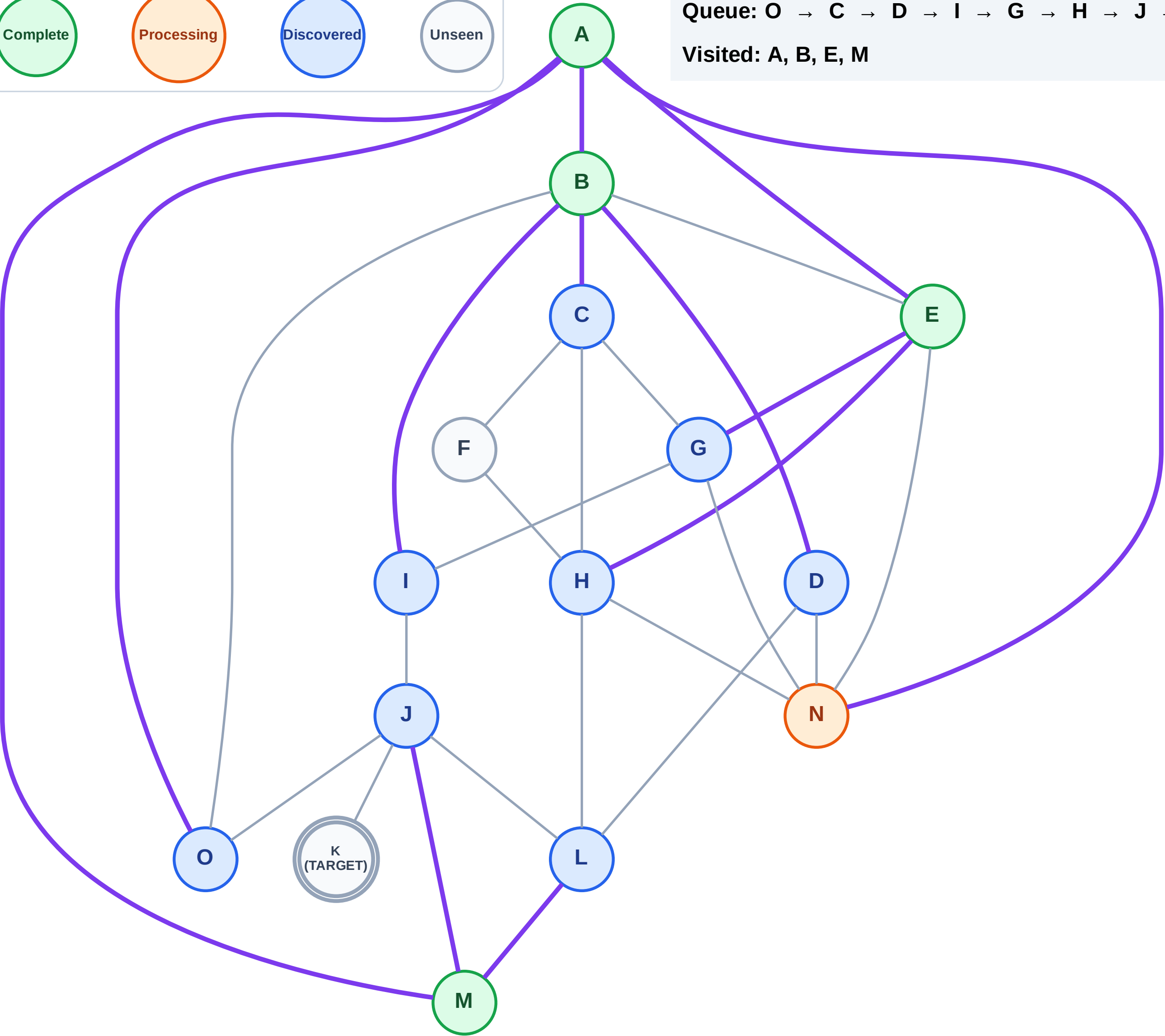
Processing

Discovered

Unseen

Queue: O → C → D → I → G → H → J → L

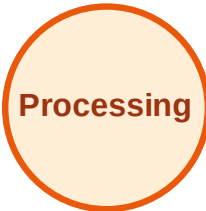
Visited: A, B, E, M



BFS Traversal

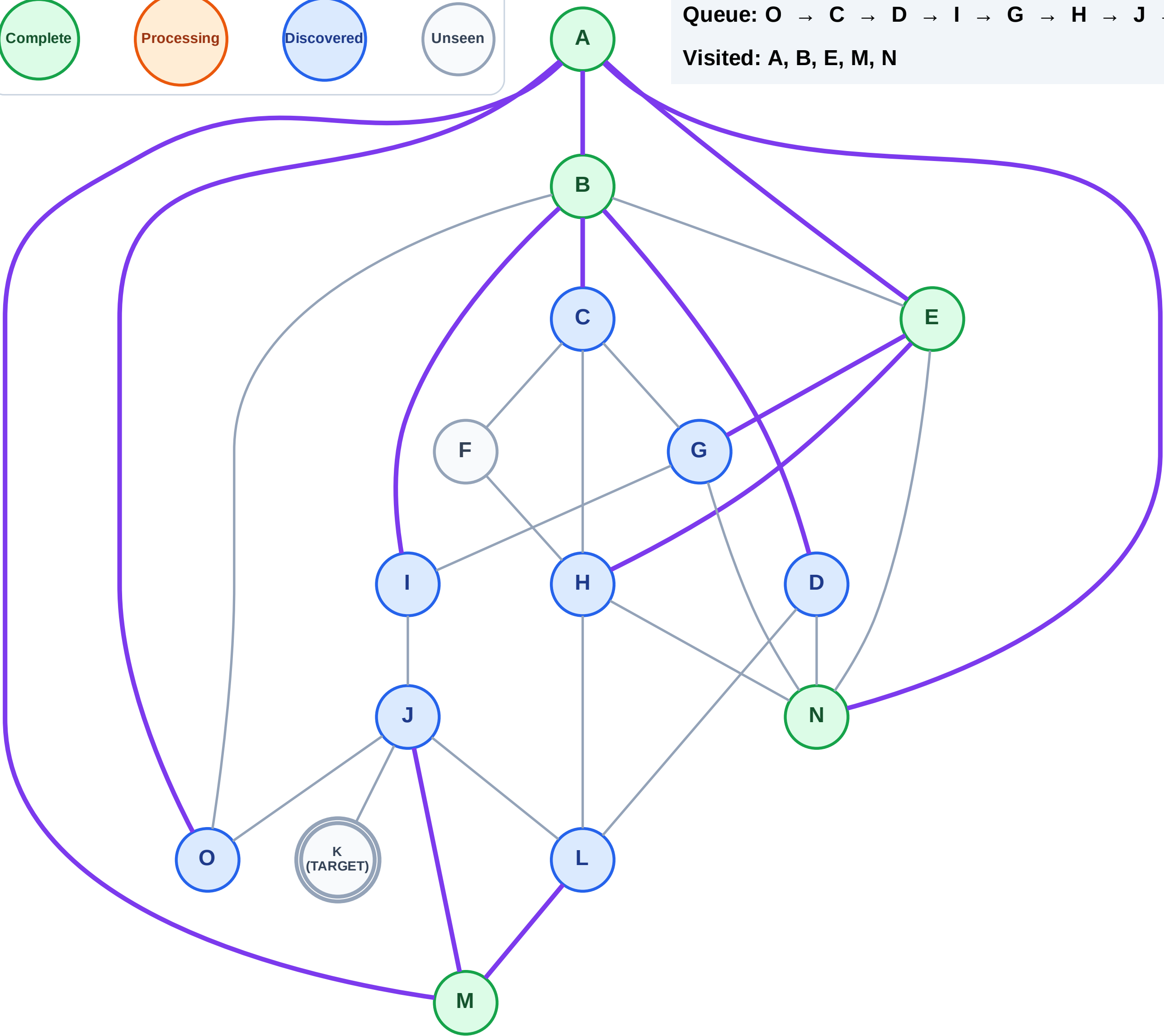
Step 11: No new neighbors from N

Legend



Queue: O → C → D → I → G → H → J → L

Visited: A, B, E, M, N

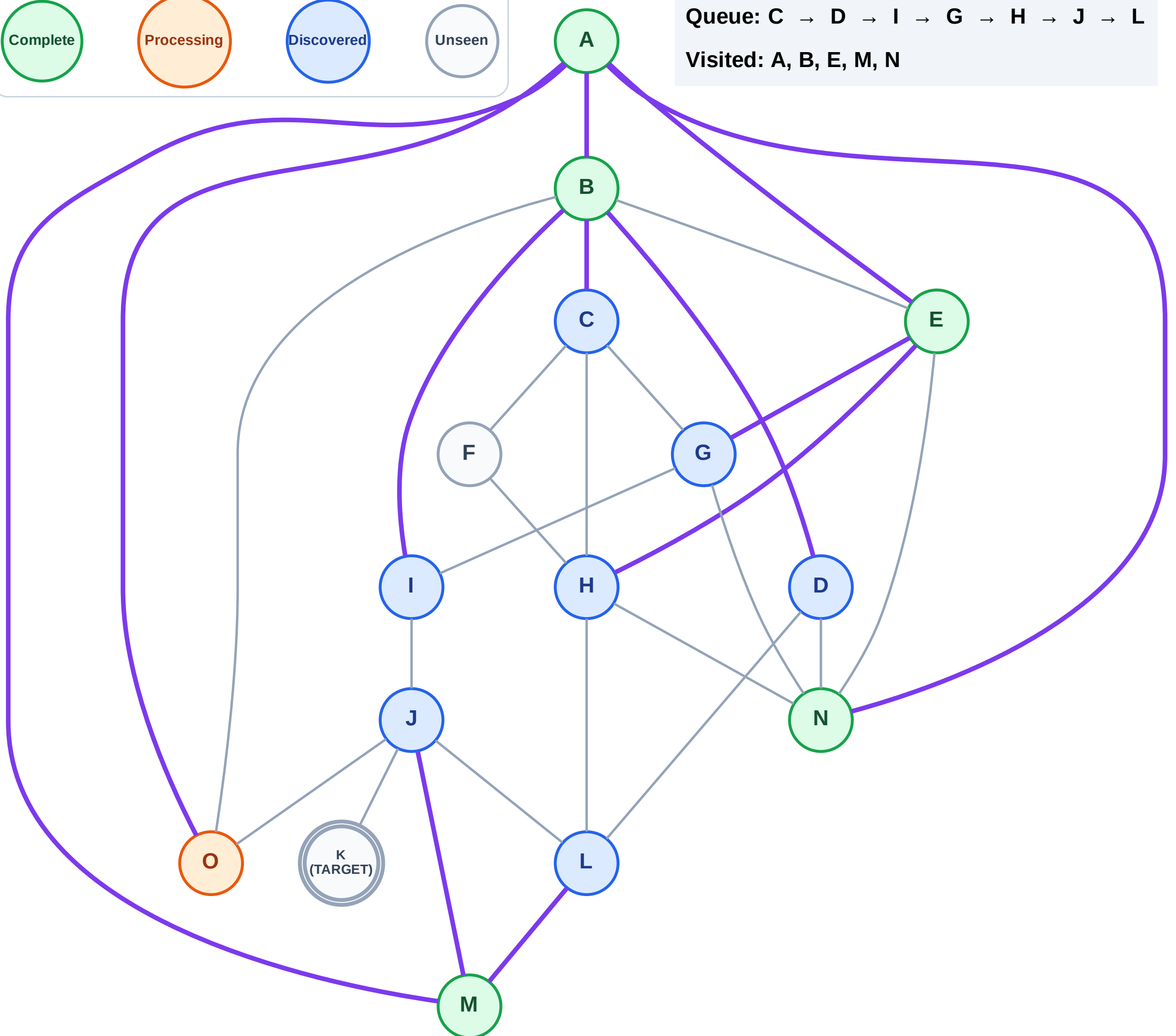


Step 12: Process node O

Legend

Unseen

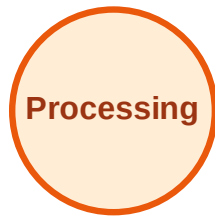
Visited: A, B, E, M, N



BFS Traversal

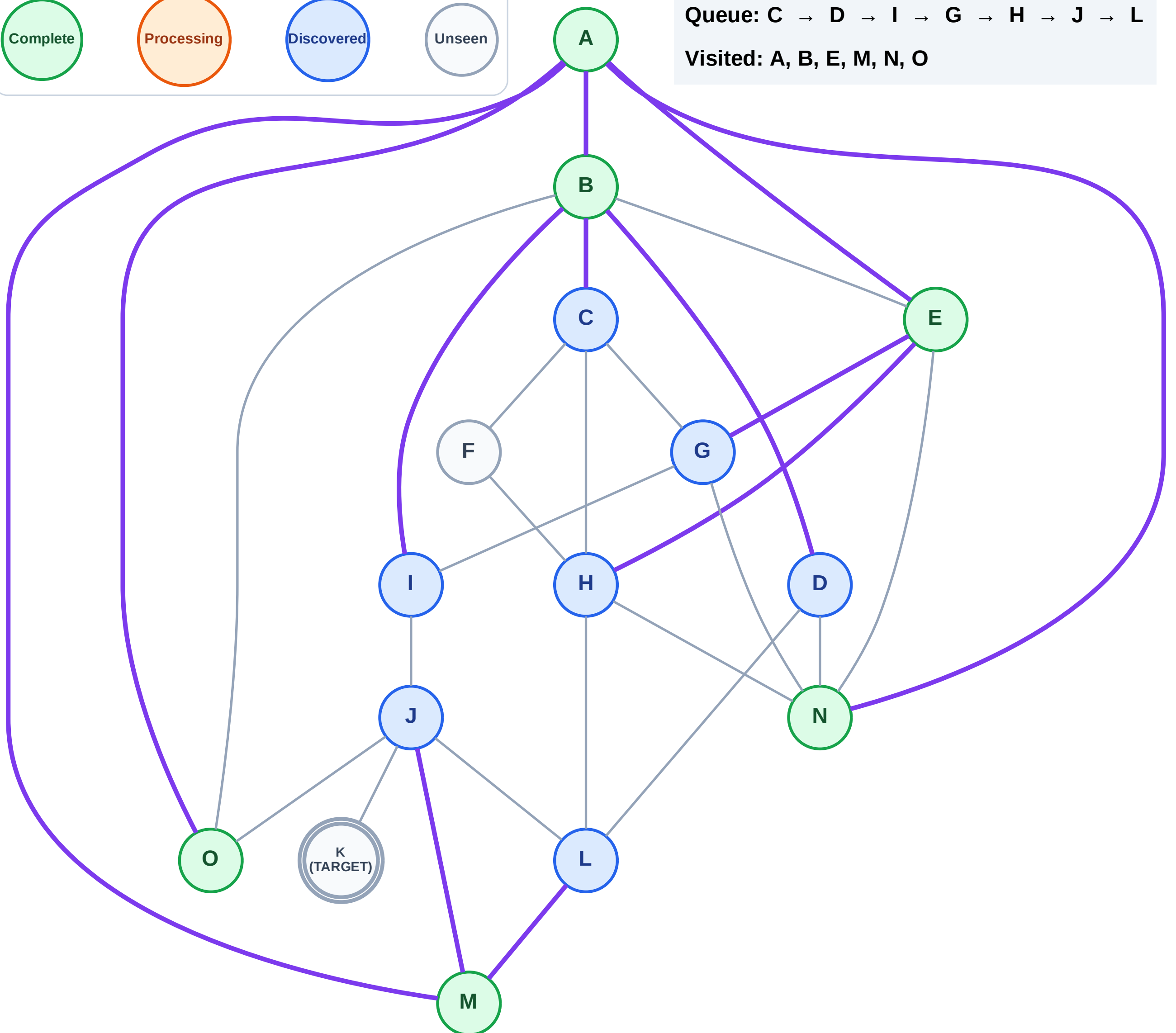
Step 13: No new neighbors from O

Legend



Queue: C → D → I → G → H → J → L

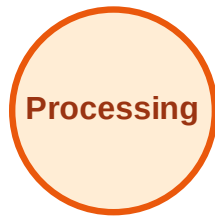
Visited: A, B, E, M, N, O



BFS Traversal

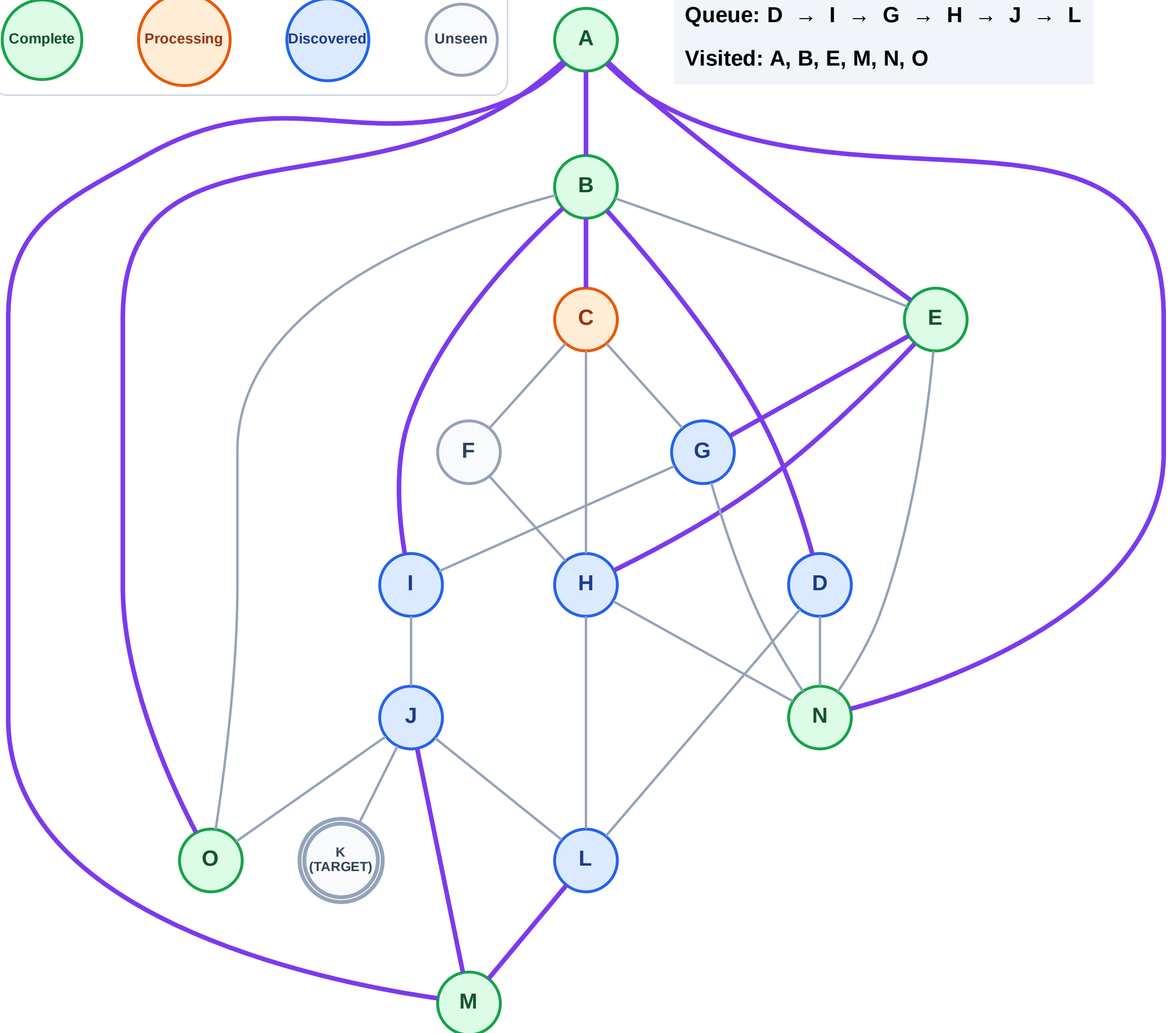
Step 14: Process node C

Legend



Queue: D → I → G → H → J → L

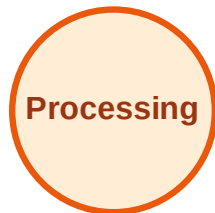
Visited: A, B, E, M, N, O



BFS Traversal

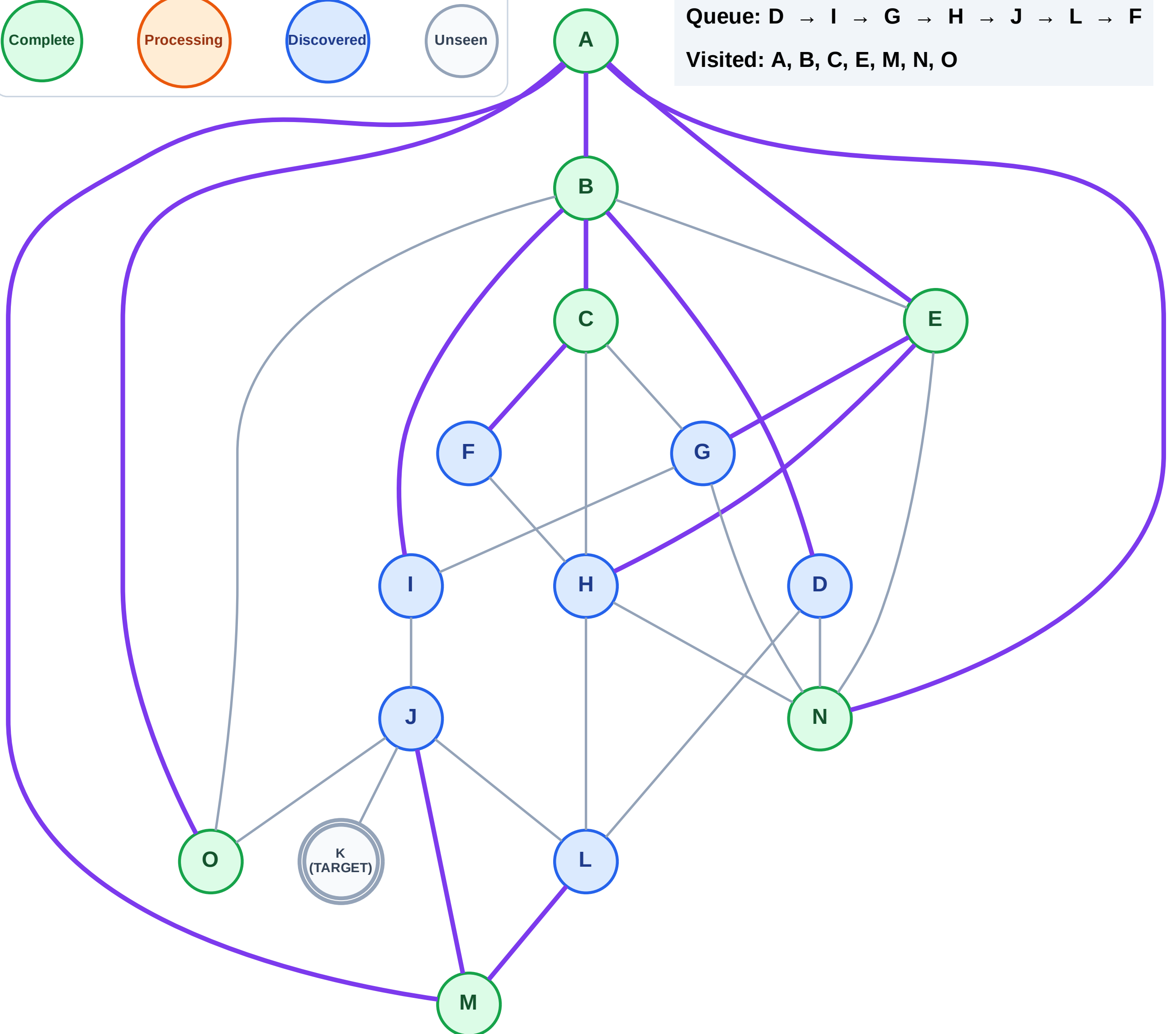
Step 15: Discover F from C

Legend



Queue: D → I → G → H → J → L → F

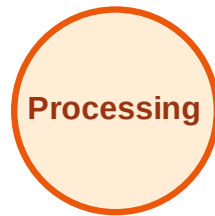
Visited: A, B, C, E, M, N, O



BFS Traversal

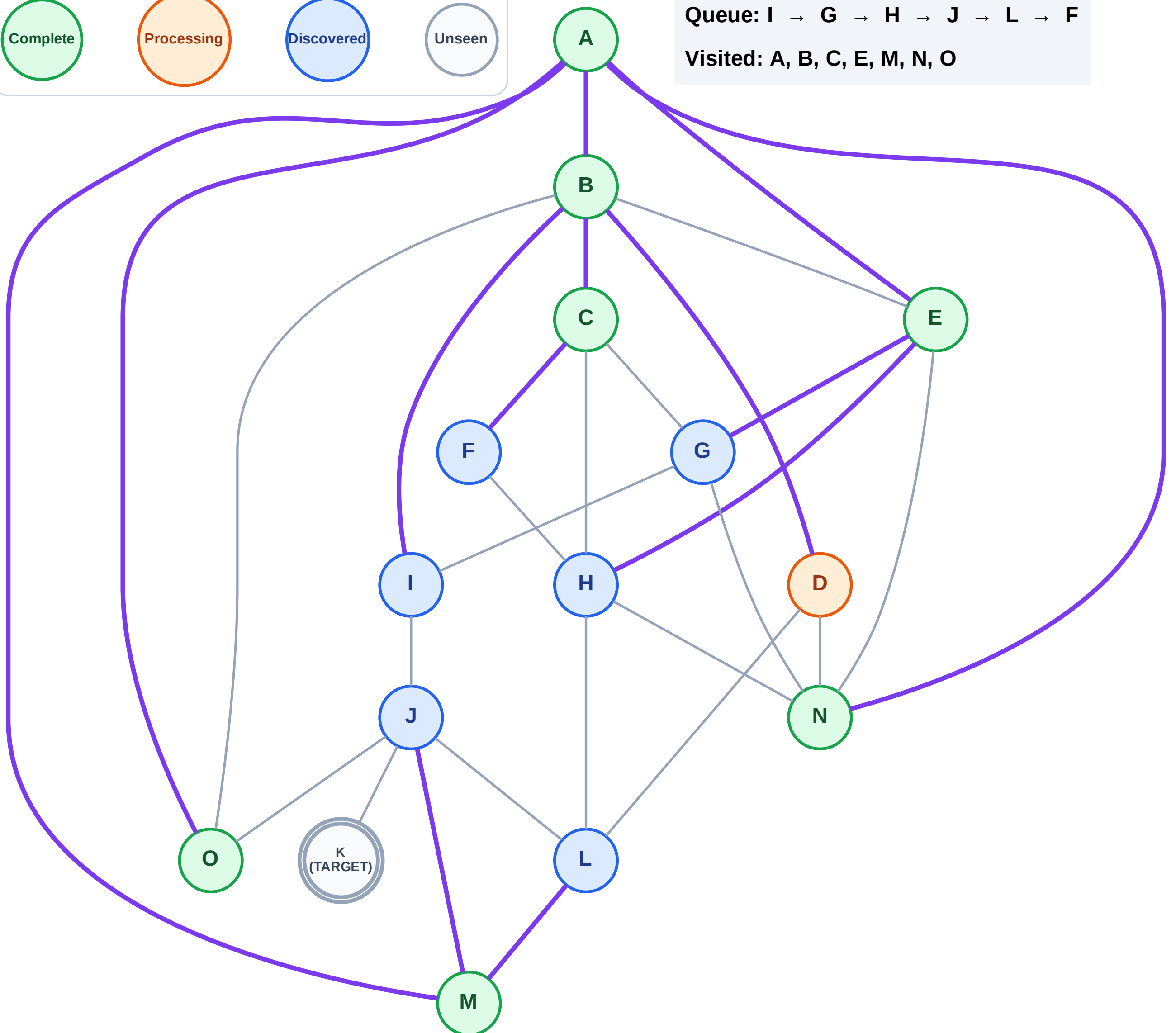
Step 16: Process node D

Legend



Queue: I → G → H → J → L → F

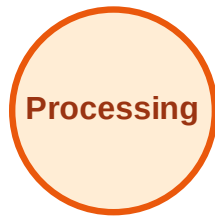
Visited: A, B, C, E, M, N, O



BFS Traversal

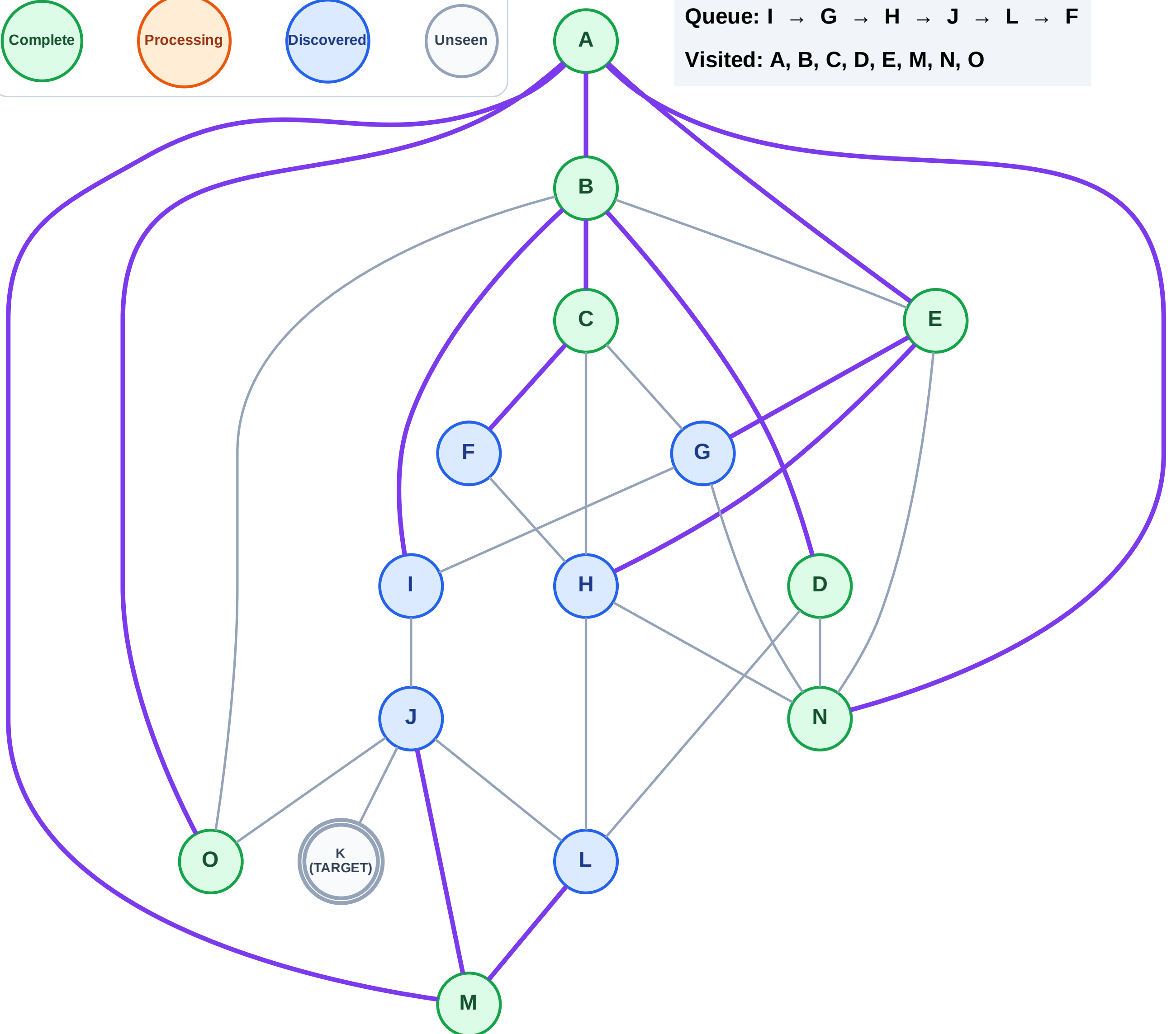
Step 17: No new neighbors from D

Legend



Queue: I → G → H → J → L → F

Visited: A, B, C, D, E, M, N, O



BFS Traversal

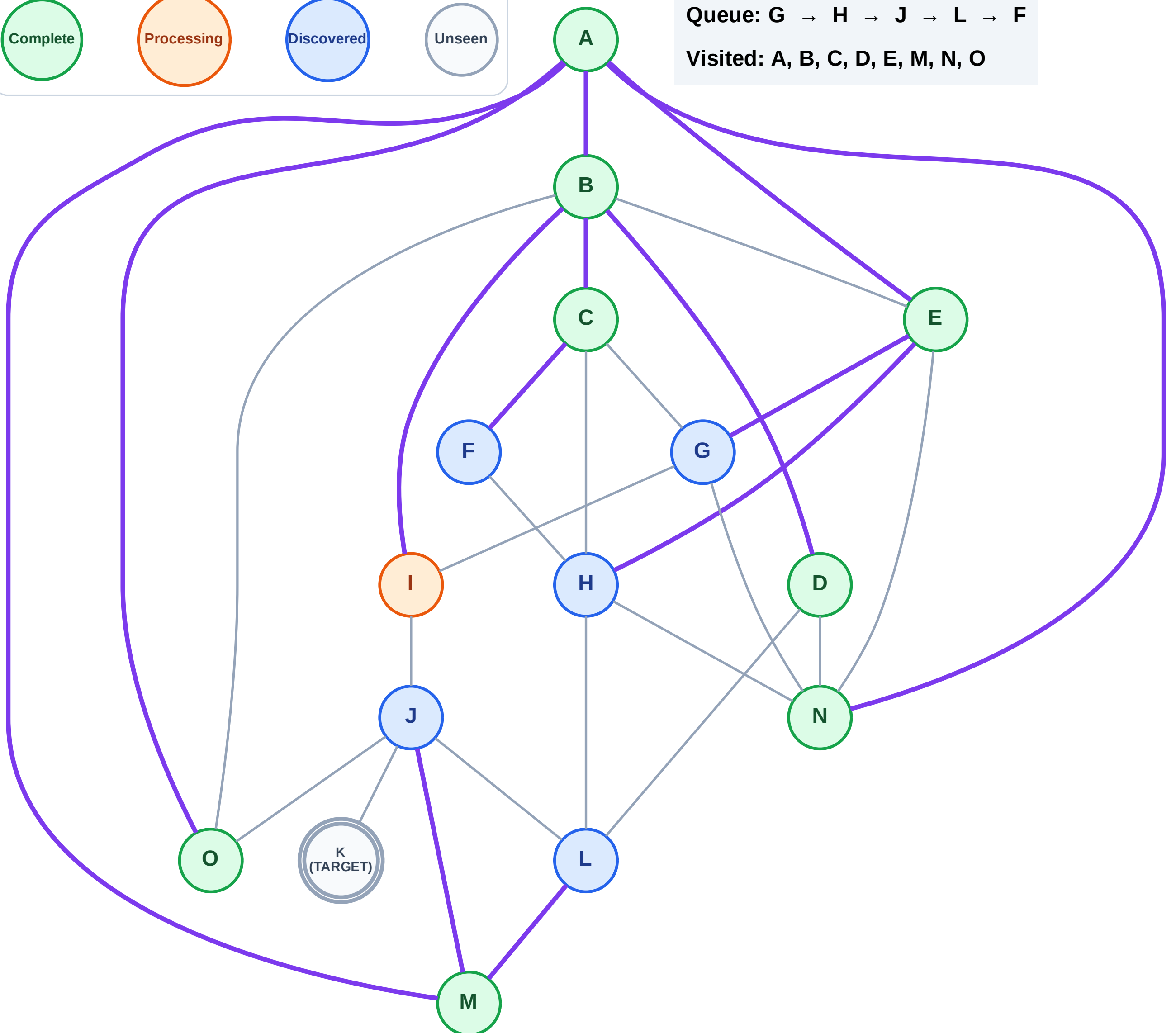
Step 18: Process node I

Legend



Queue: G → H → J → L → F

Visited: A, B, C, D, E, M, N, O



BFS Traversal

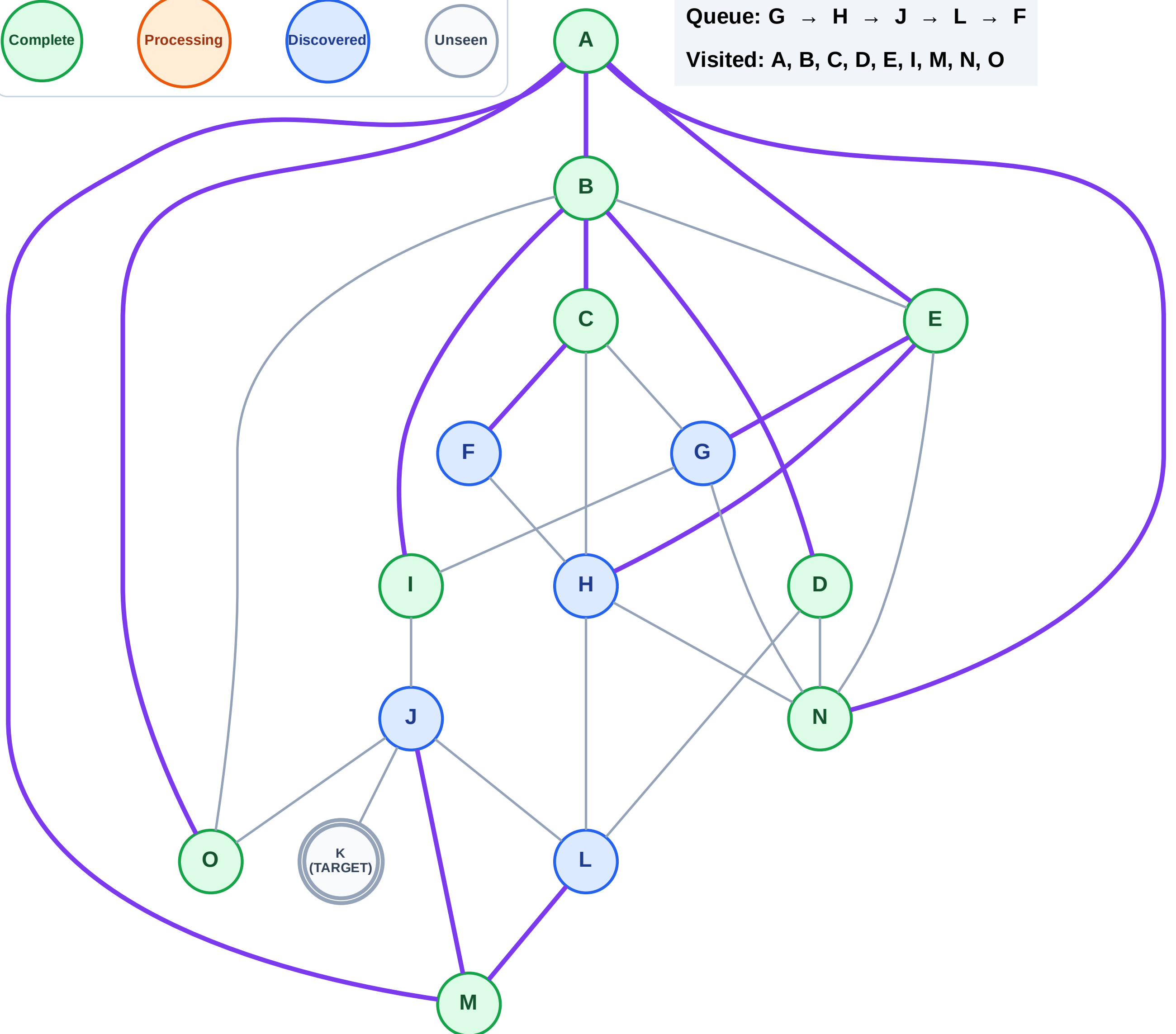
Step 19: No new neighbors from I

Legend

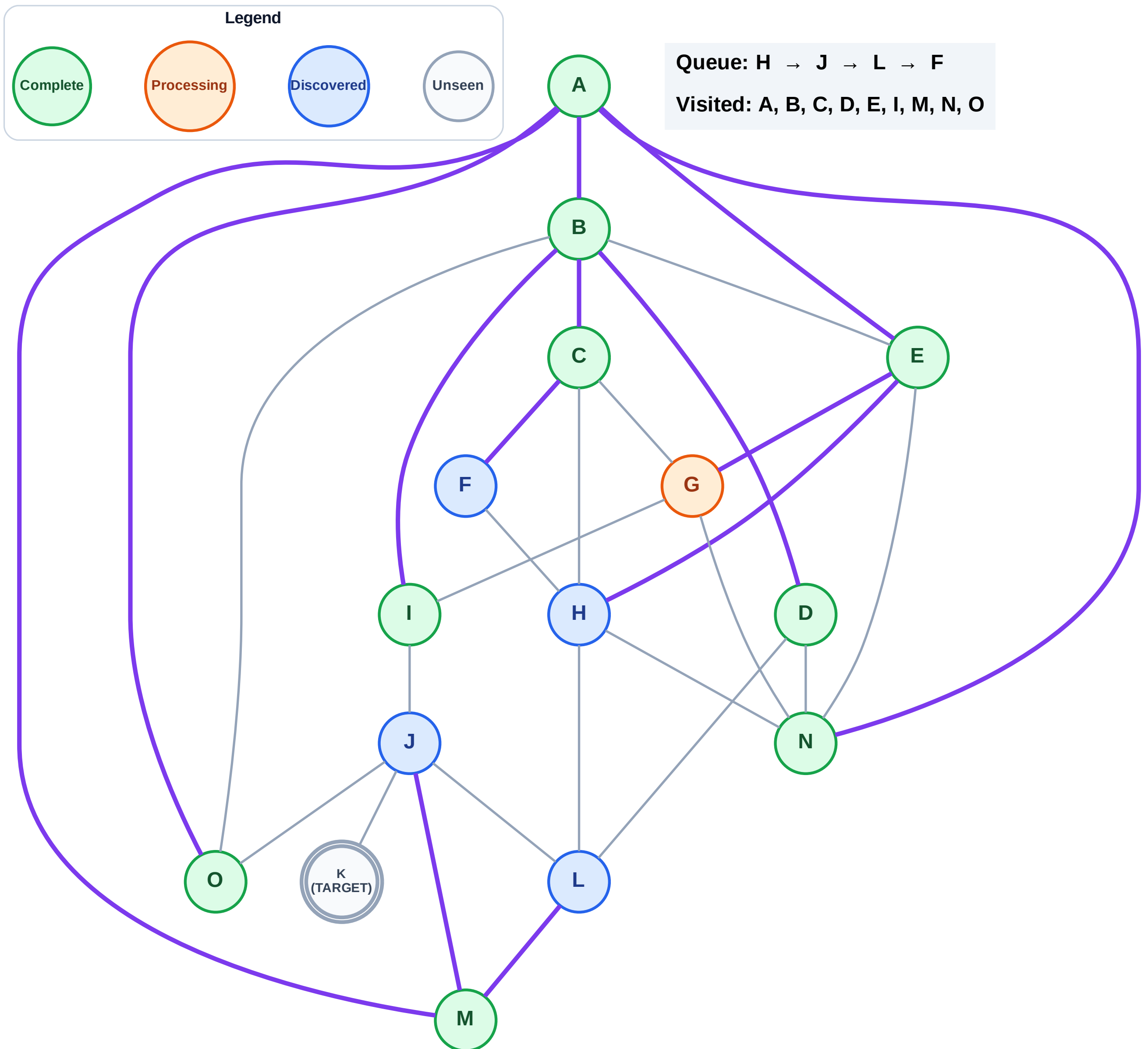


Queue: G → H → J → L → F

Visited: A, B, C, D, E, I, M, N, O



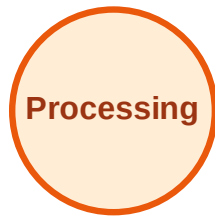
Step 20: Process node G



BFS Traversal

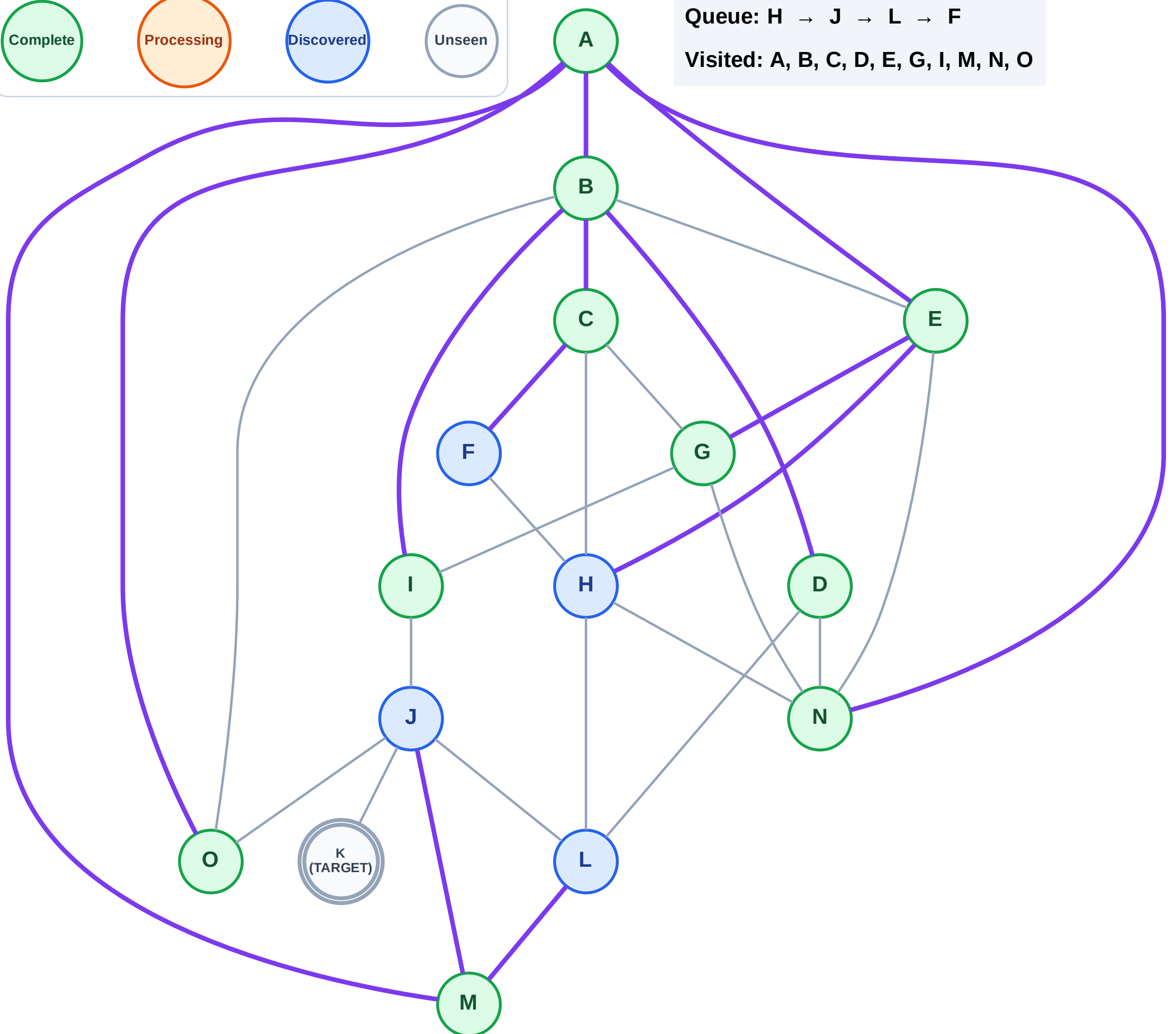
Step 21: No new neighbors from G

Legend



Queue: H → J → L → F

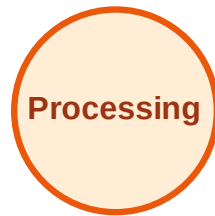
Visited: A, B, C, D, E, G, I, M, N, O



BFS Traversal

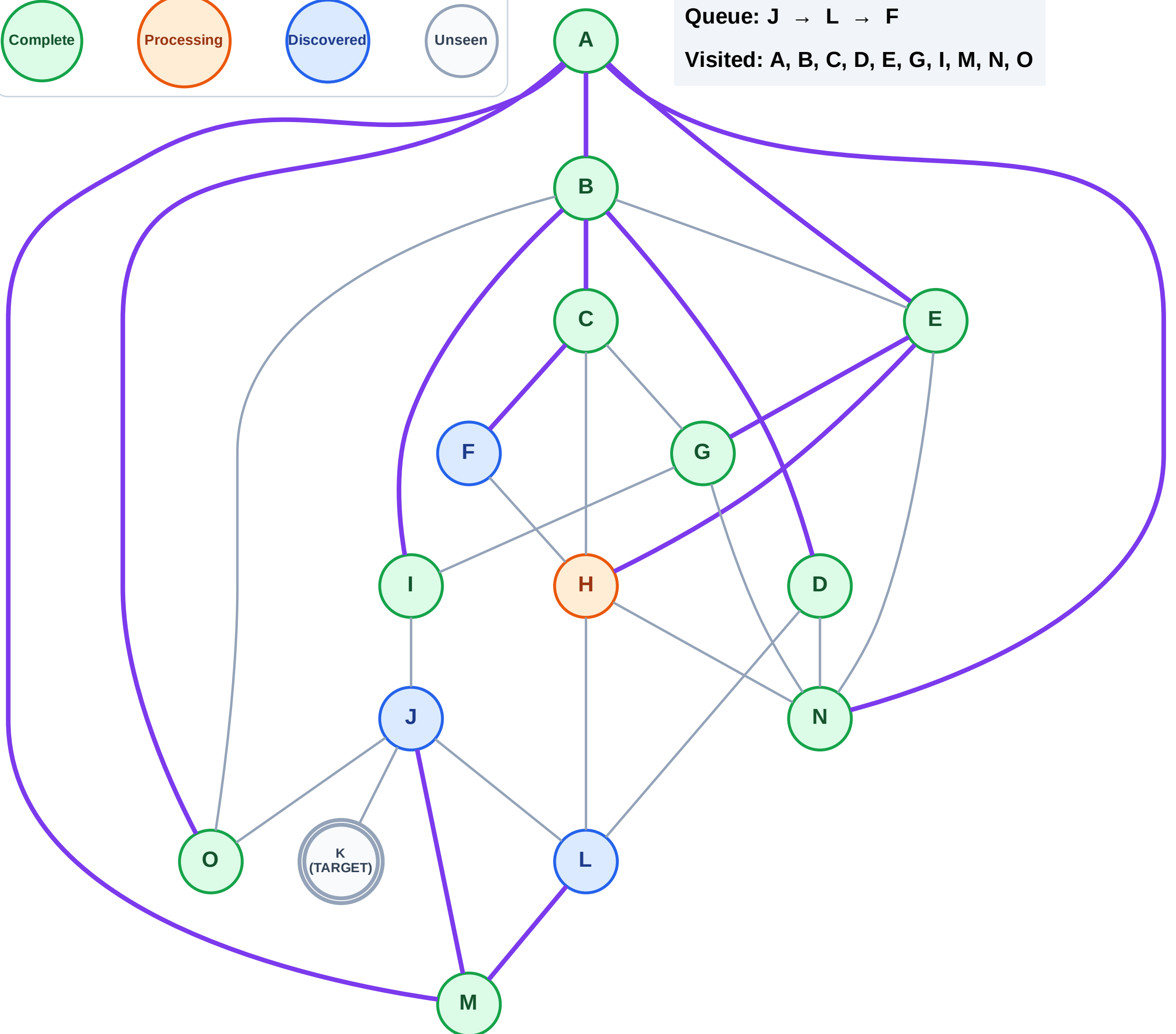
Step 22: Process node H

Legend



Queue: J → L → F

Visited: A, B, C, D, E, G, I, M, N, O



BFS Traversal

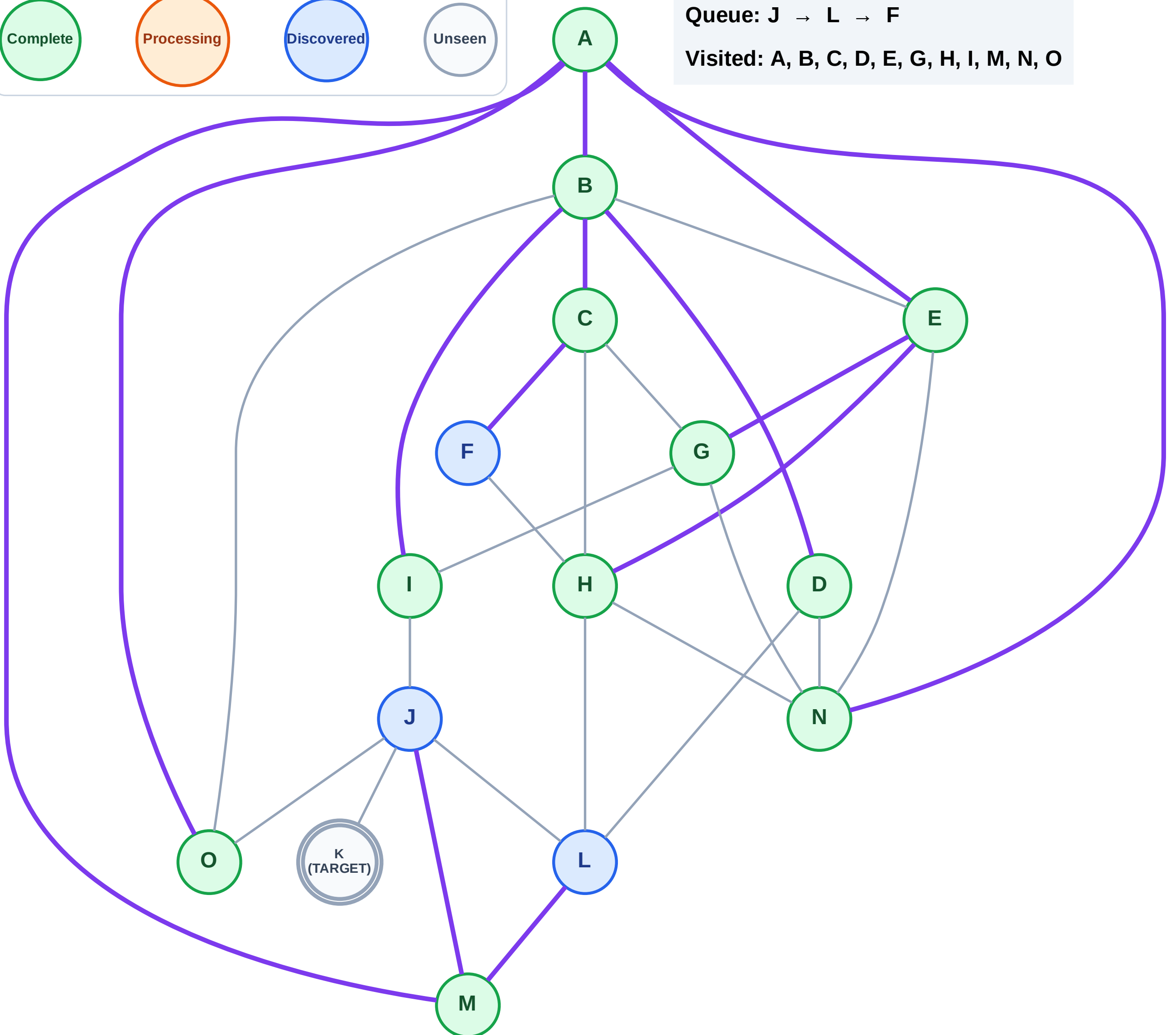
Step 23: No new neighbors from H

Legend



Queue: J → L → F

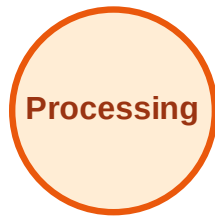
Visited: A, B, C, D, E, G, H, I, M, N, O



BFS Traversal

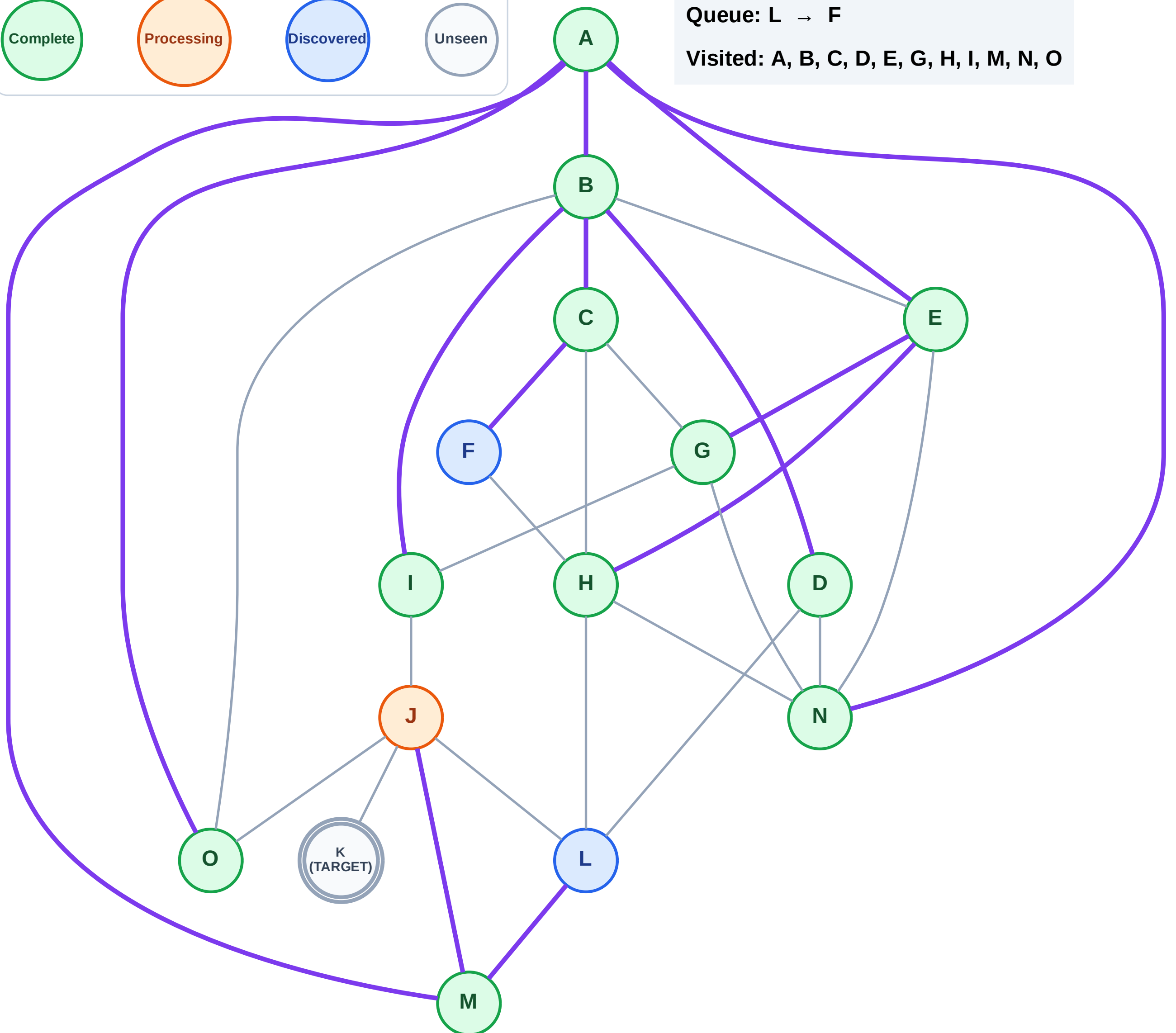
Step 24: Process node J

Legend



Queue: L → F

Visited: A, B, C, D, E, G, H, I, M, N, O



BFS Traversal

Step 25: TARGET NODE K FOUND (SHORT-CIRCUITED)! Path (3 hops): A → M → J → K

Legend

Complete

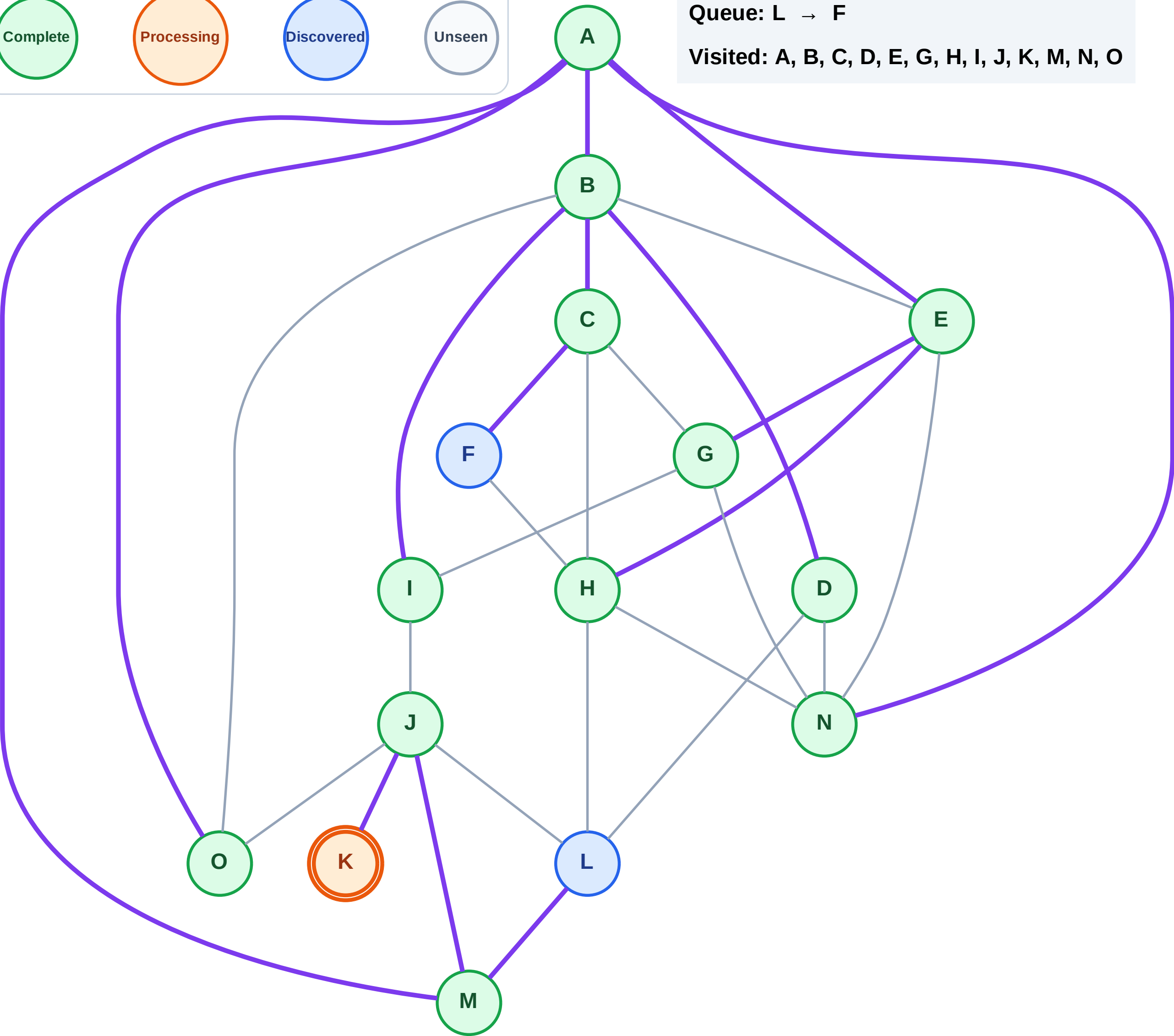
Processing

Discovered

Unseen

Queue: L → F

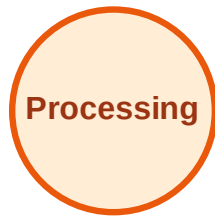
Visited: A, B, C, D, E, G, H, I, J, K, M, N, O



DFS Traversal

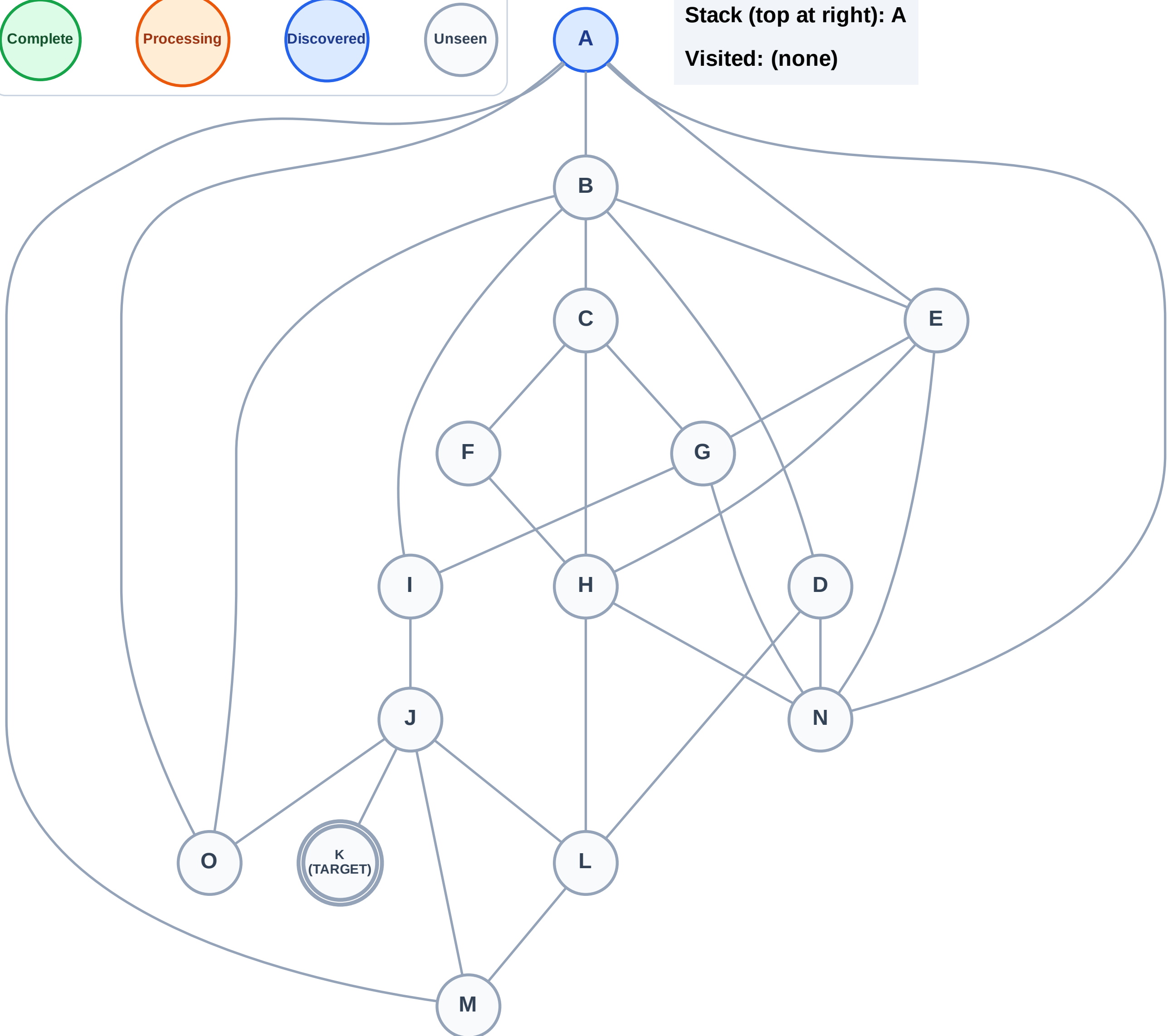
Step 1: Start at A (Seeking target node K (Eager Short-Circuit))

Legend



Stack (top at right): A

Visited: (none)



DFS Traversal

Step 2: Process node A

Legend

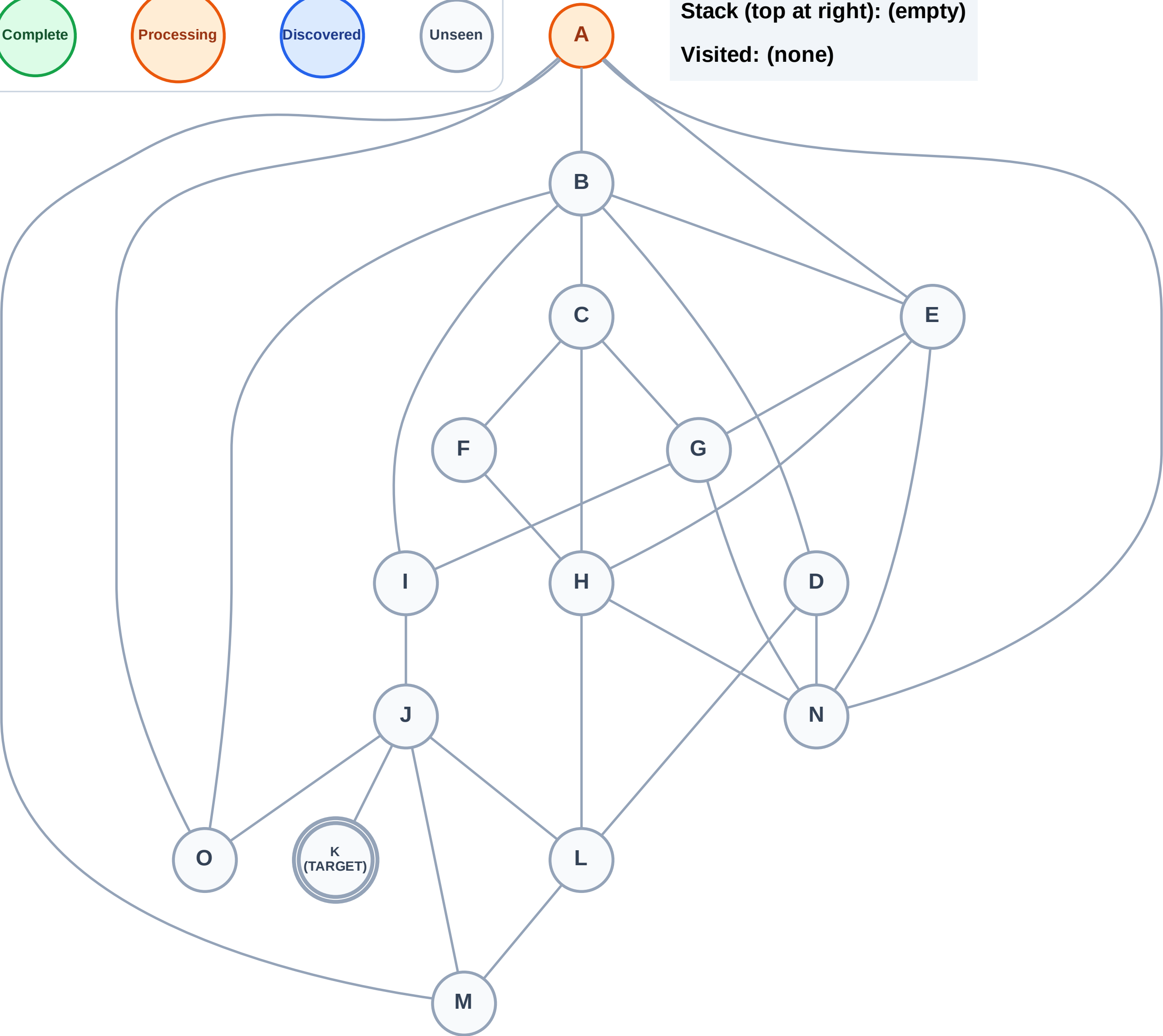
Complete

Processing

Discovered

Unseen

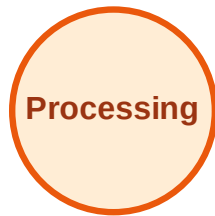
Stack (top at right): (empty)
Visited: (none)



DFS Traversal

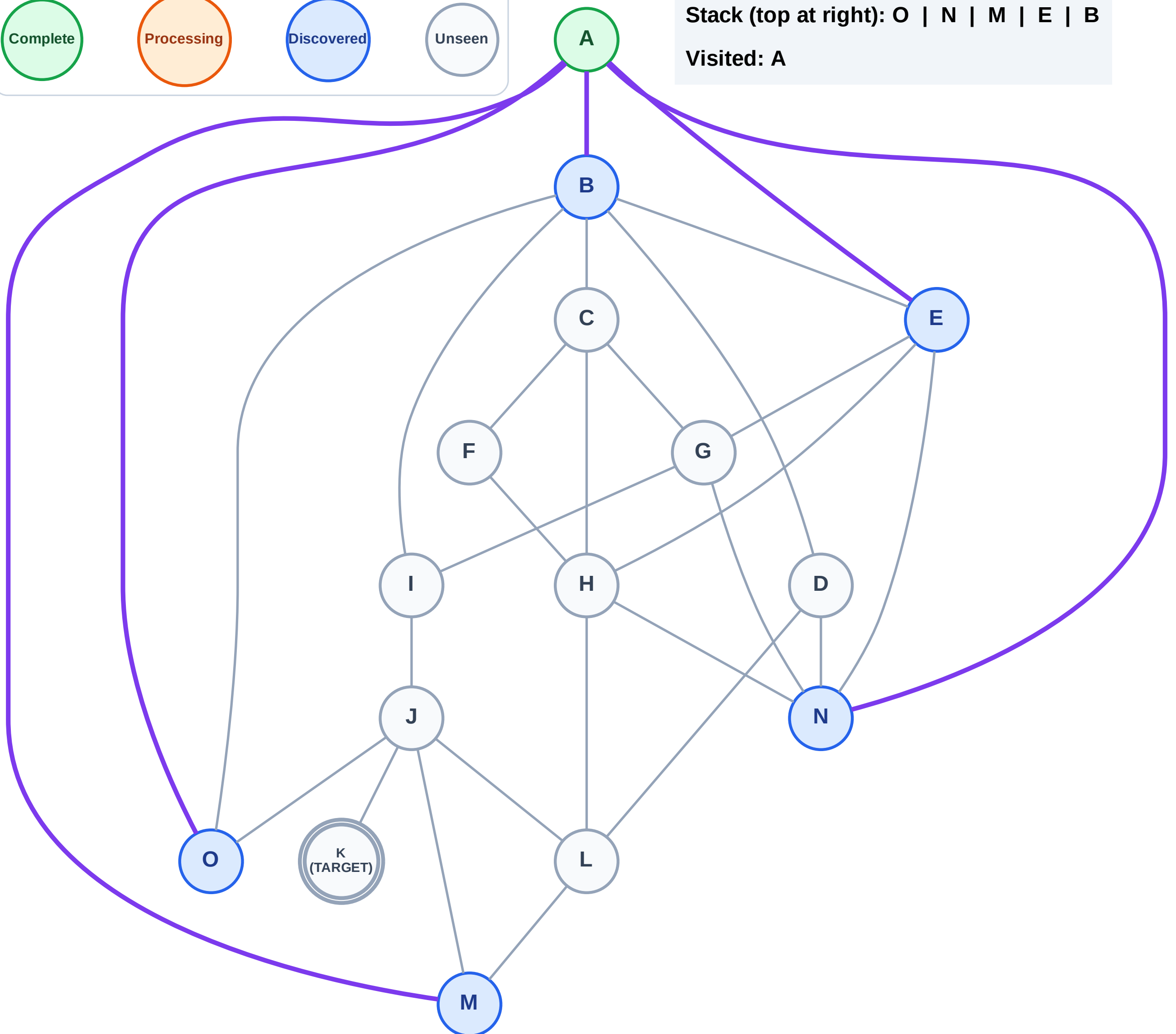
Step 3: Discover O, N, M, E, B from A

Legend



Stack (top at right): O | N | M | E | B

Visited: A



DFS Traversal

Step 4: Process node B

Legend

Complete

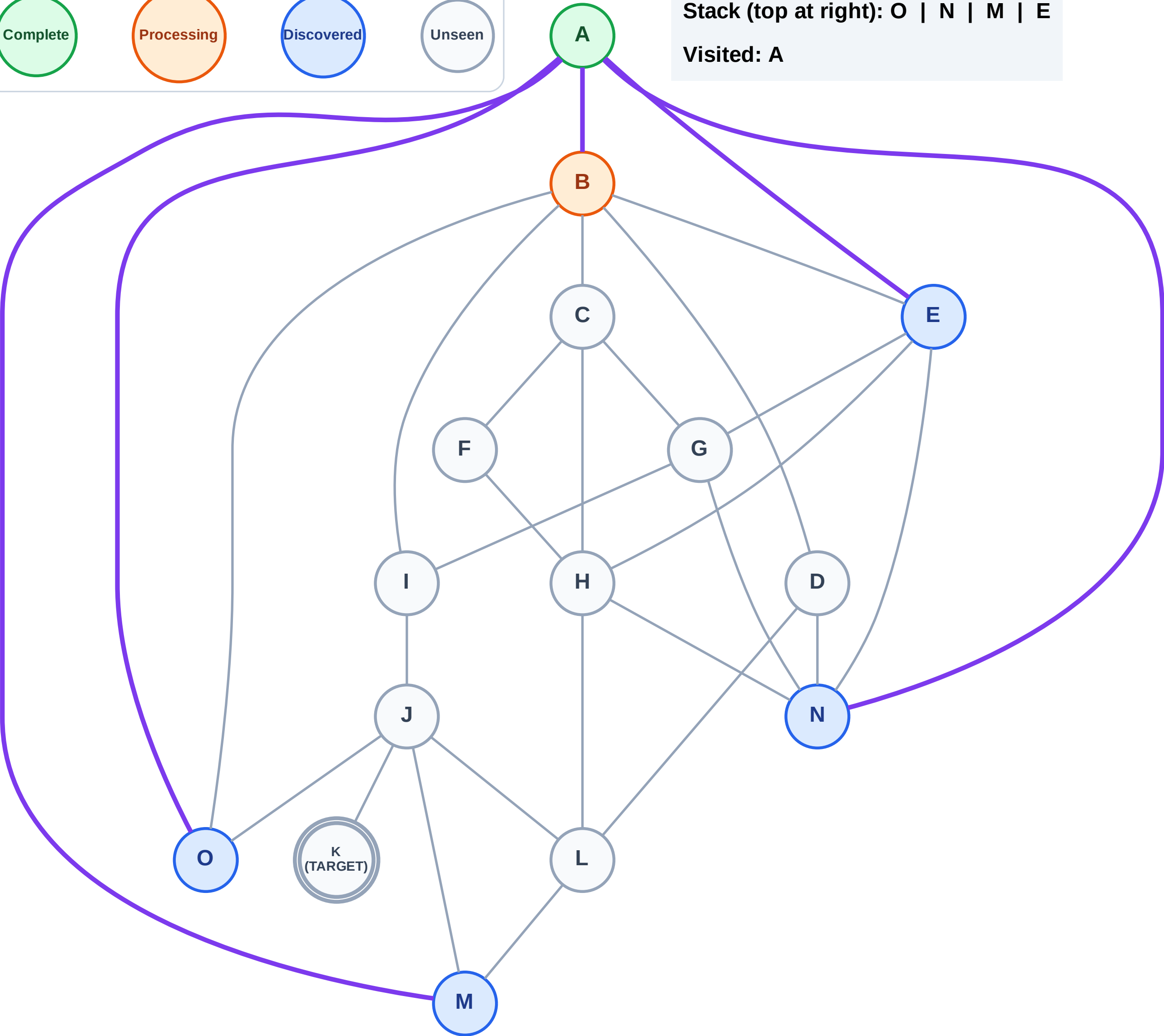
Processing

Discovered

Unseen

Stack (top at right): O | N | M | E

Visited: A



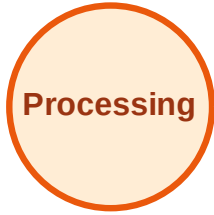
DFS Traversal

Step 5: Discover I, D, C from B

Legend



Complete



Processing



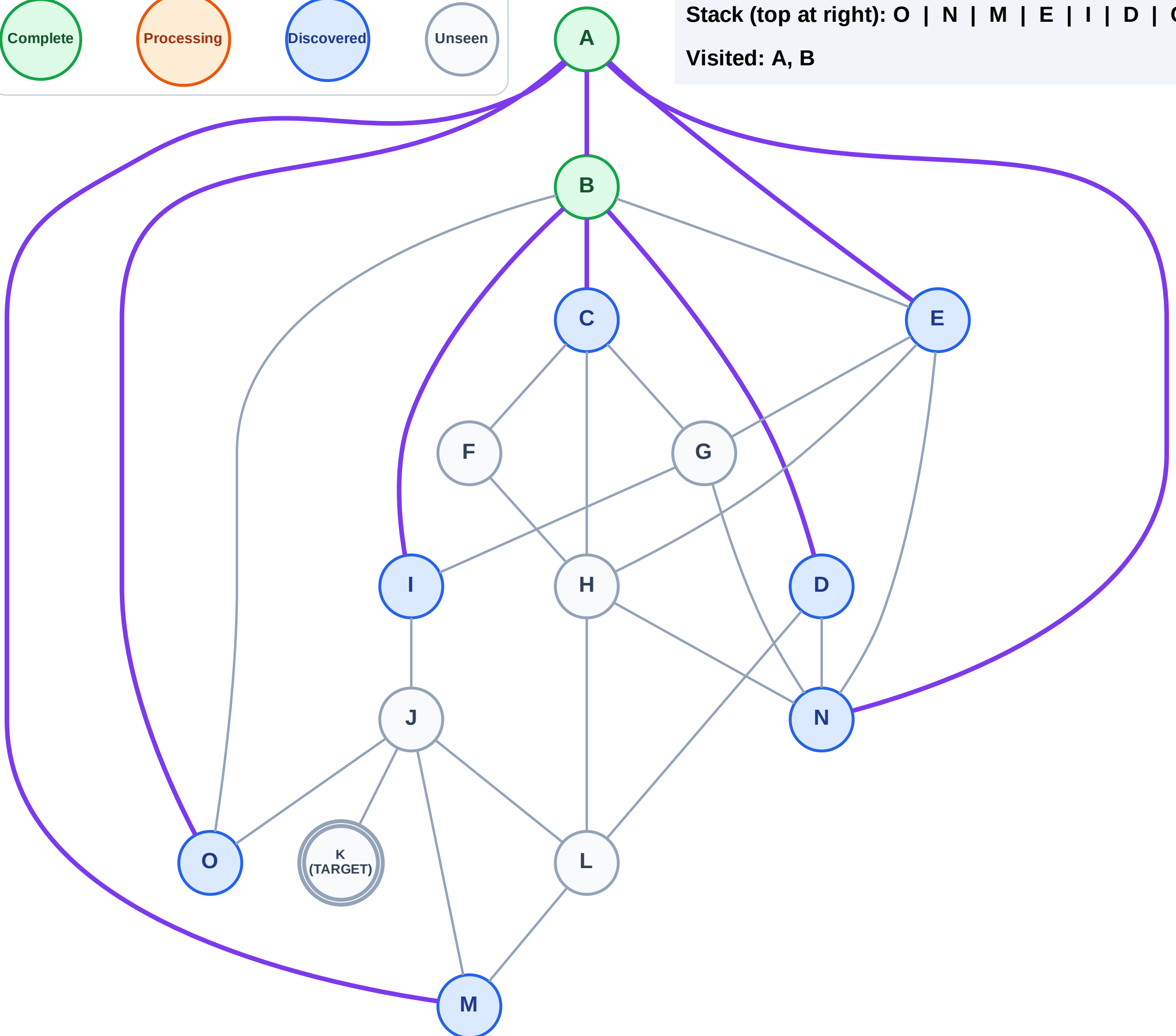
Discovered



Unseen

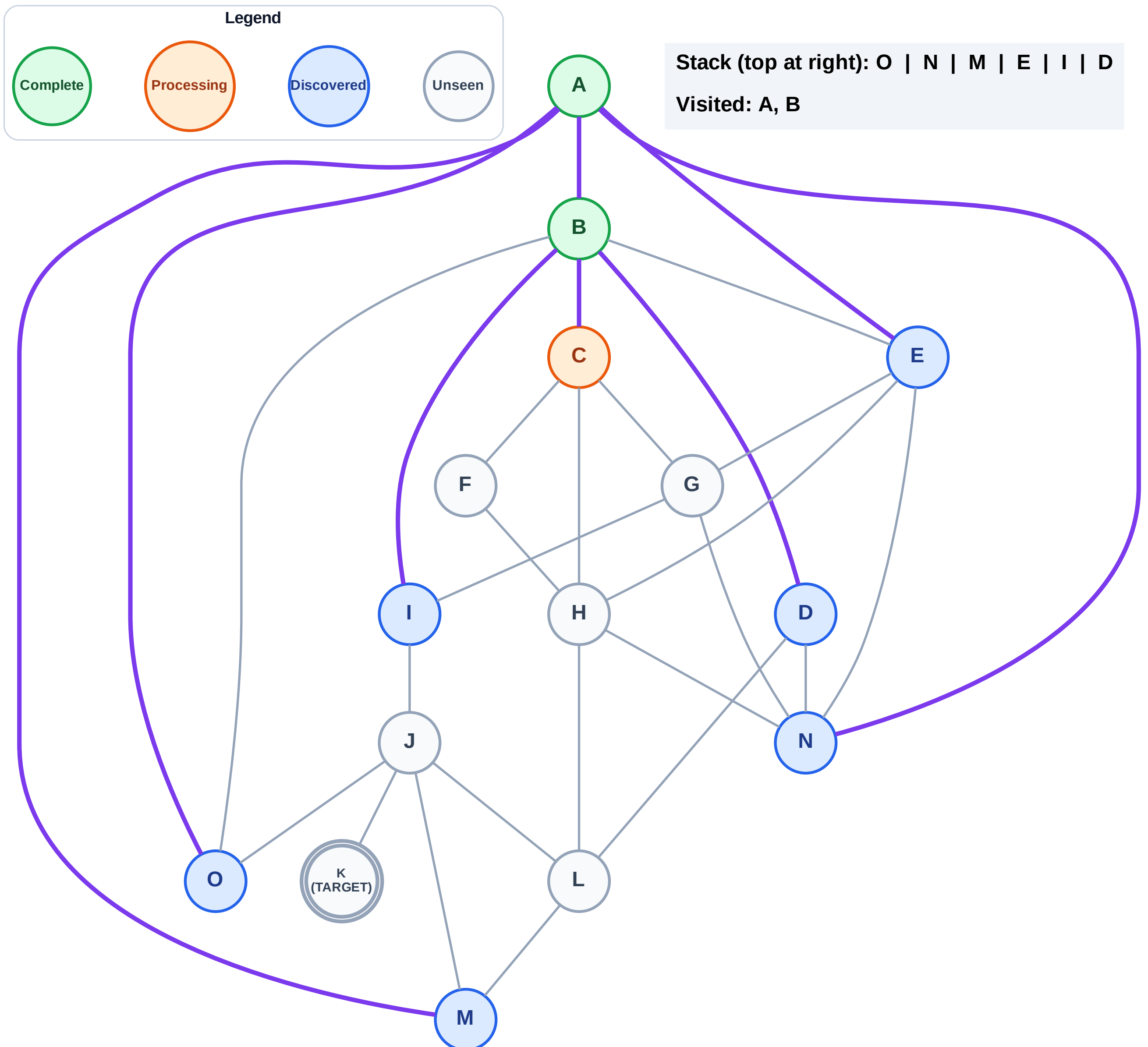
Stack (top at right): O | N | M | E | I | D | C

Visited: A, B



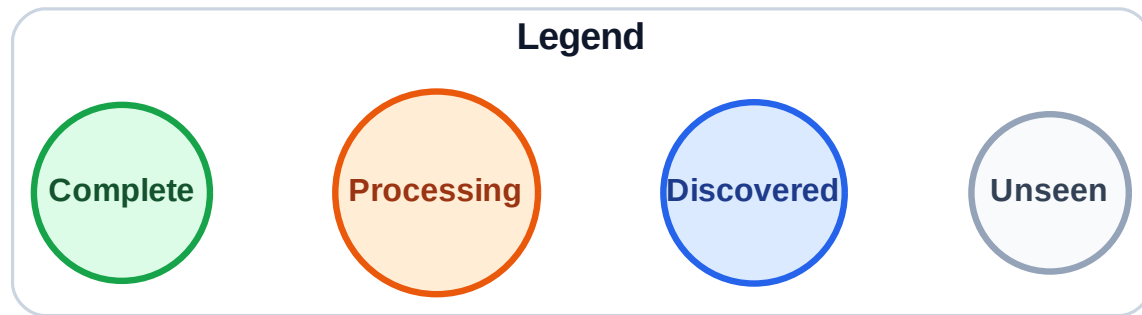
Step 6: Process node C

Step 6: Process node C



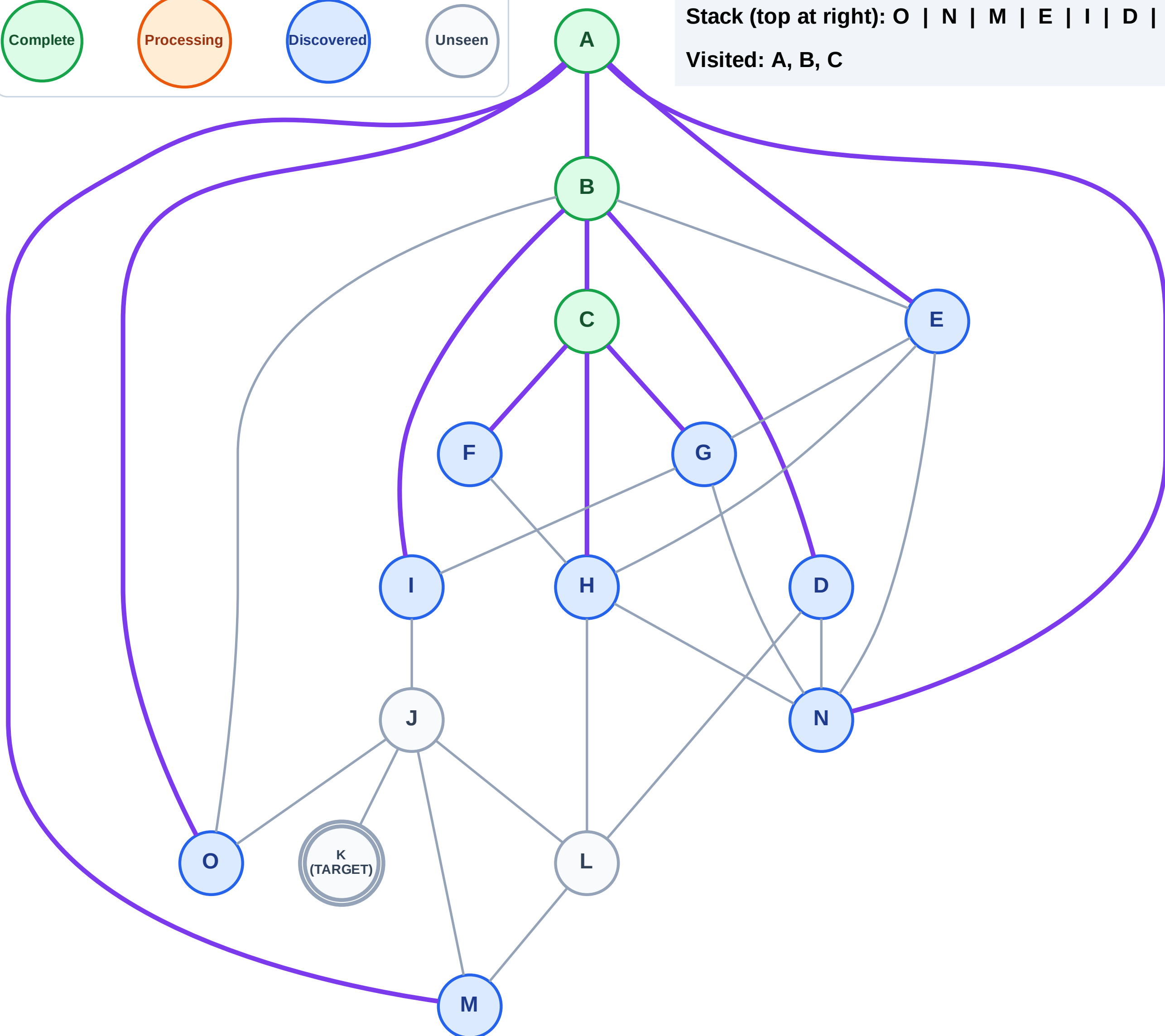
DFS Traversal

Step 7: Discover H, G, F from C



Stack (top at right): O | N | M | E | I | D | H | G | F

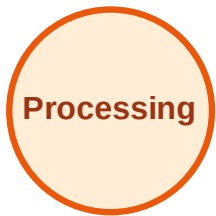
Visited: A, B, C



DFS Traversal

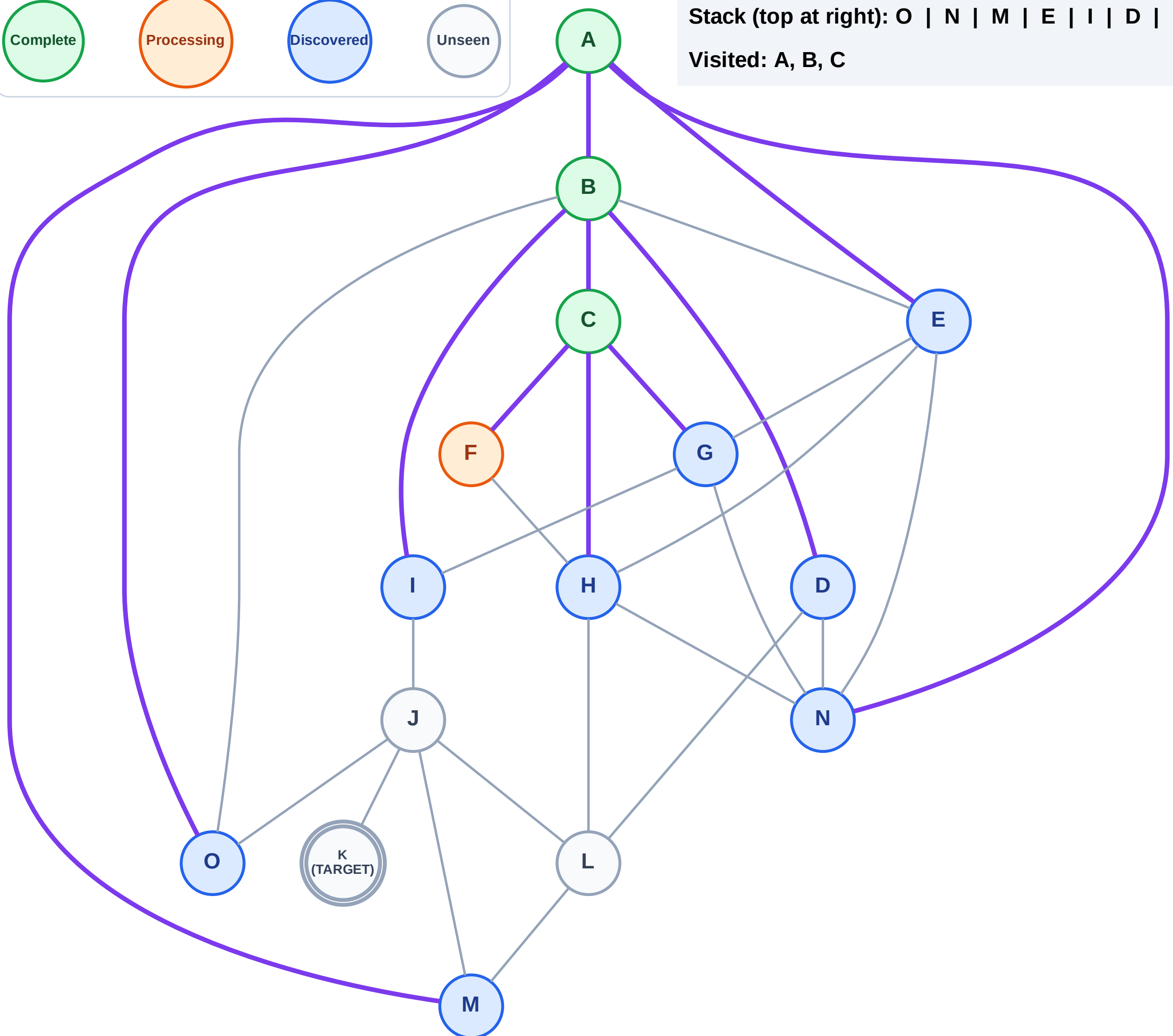
Step 8: Process node F

Legend



Stack (top at right): O | N | M | E | I | D | H | G

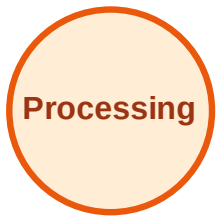
Visited: A, B, C



DFS Traversal

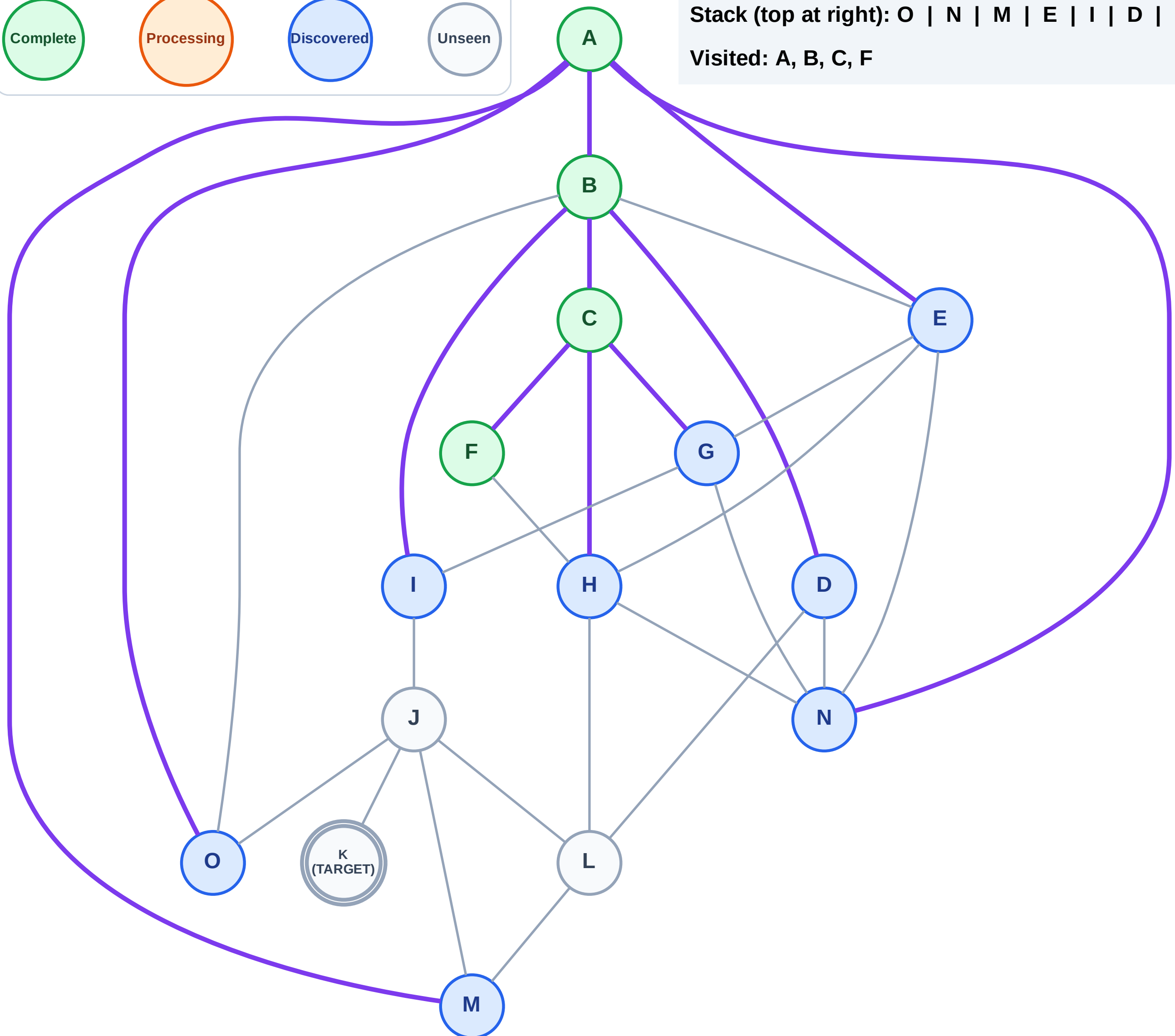
Step 9: No new neighbors from F

Legend



Stack (top at right): O | N | M | E | I | D | H | G

Visited: A, B, C, F



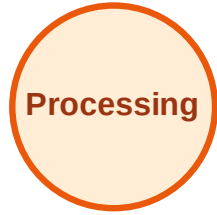
DFS Traversal

Step 10: Process node G

Legend



Complete



Processing



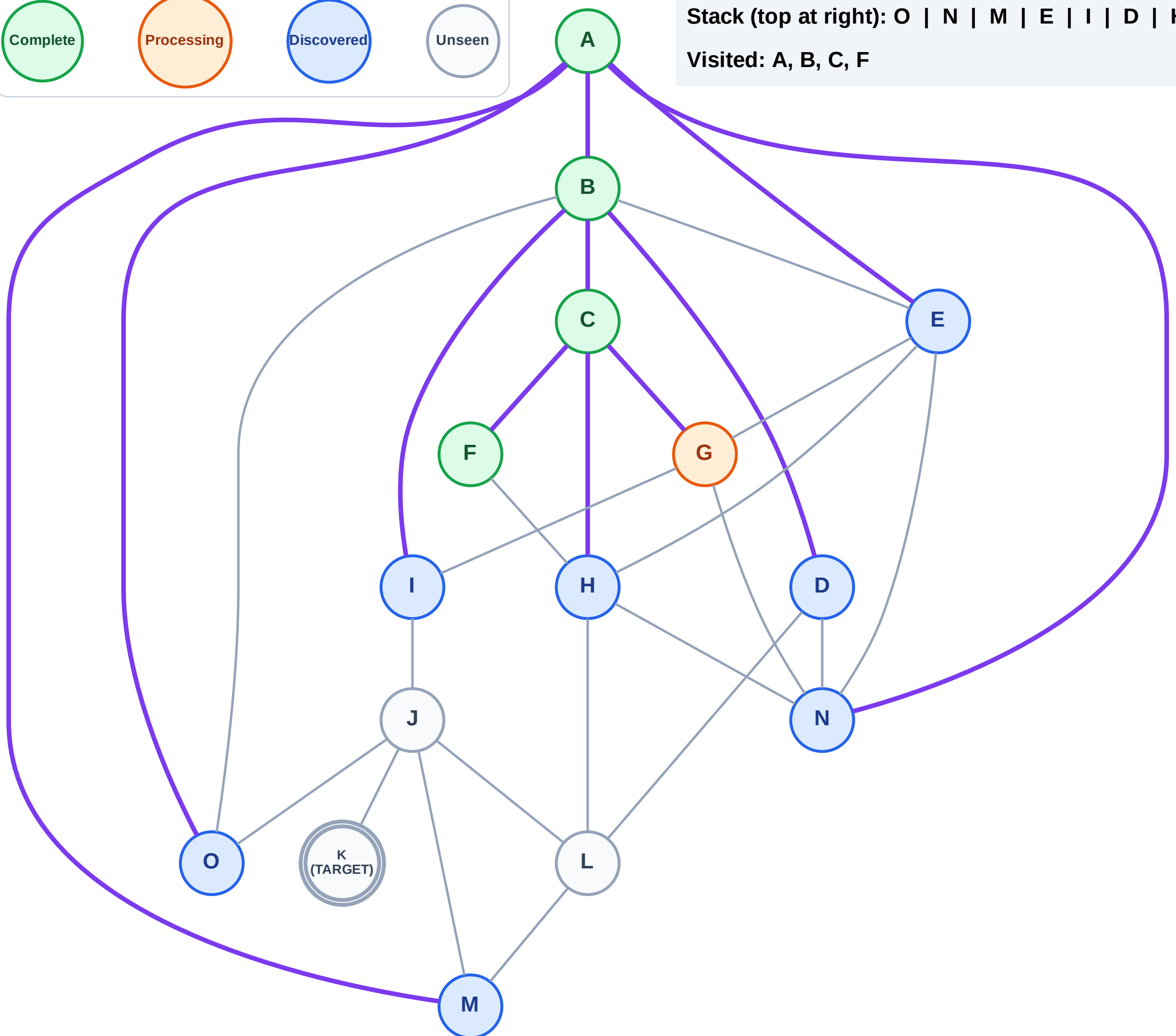
Discovered



Unseen

Stack (top at right): O | N | M | E | I | D | H

Visited: A, B, C, F



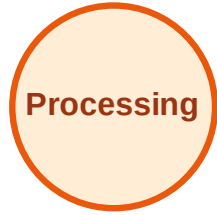
DFS Traversal

Step 11: No new neighbors from G

Legend



Complete



Processing



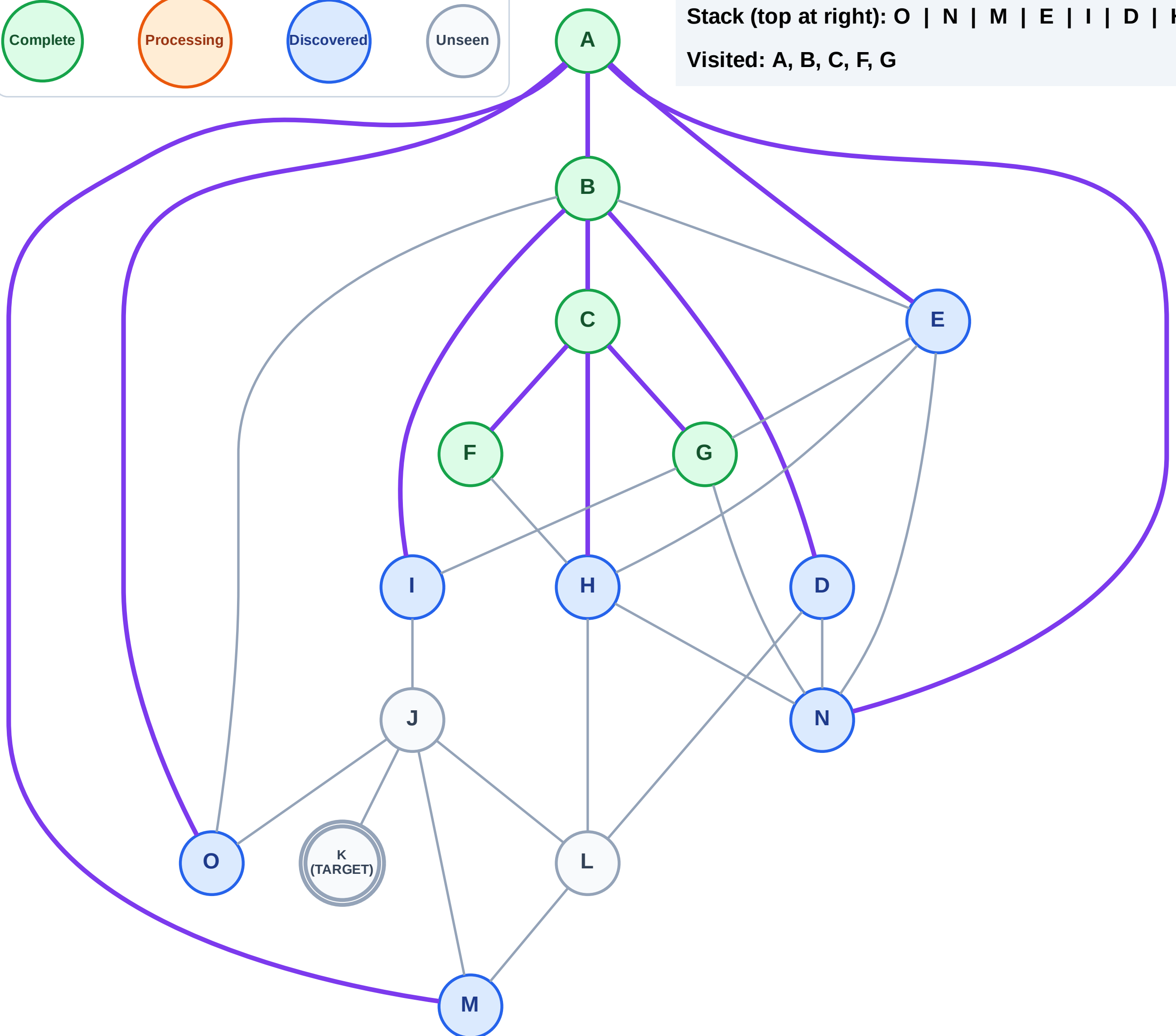
Discovered



Unseen

Stack (top at right): O | N | M | E | I | D | H

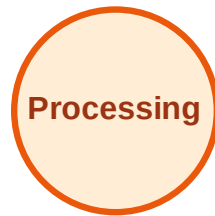
Visited: A, B, C, F, G



DFS Traversal

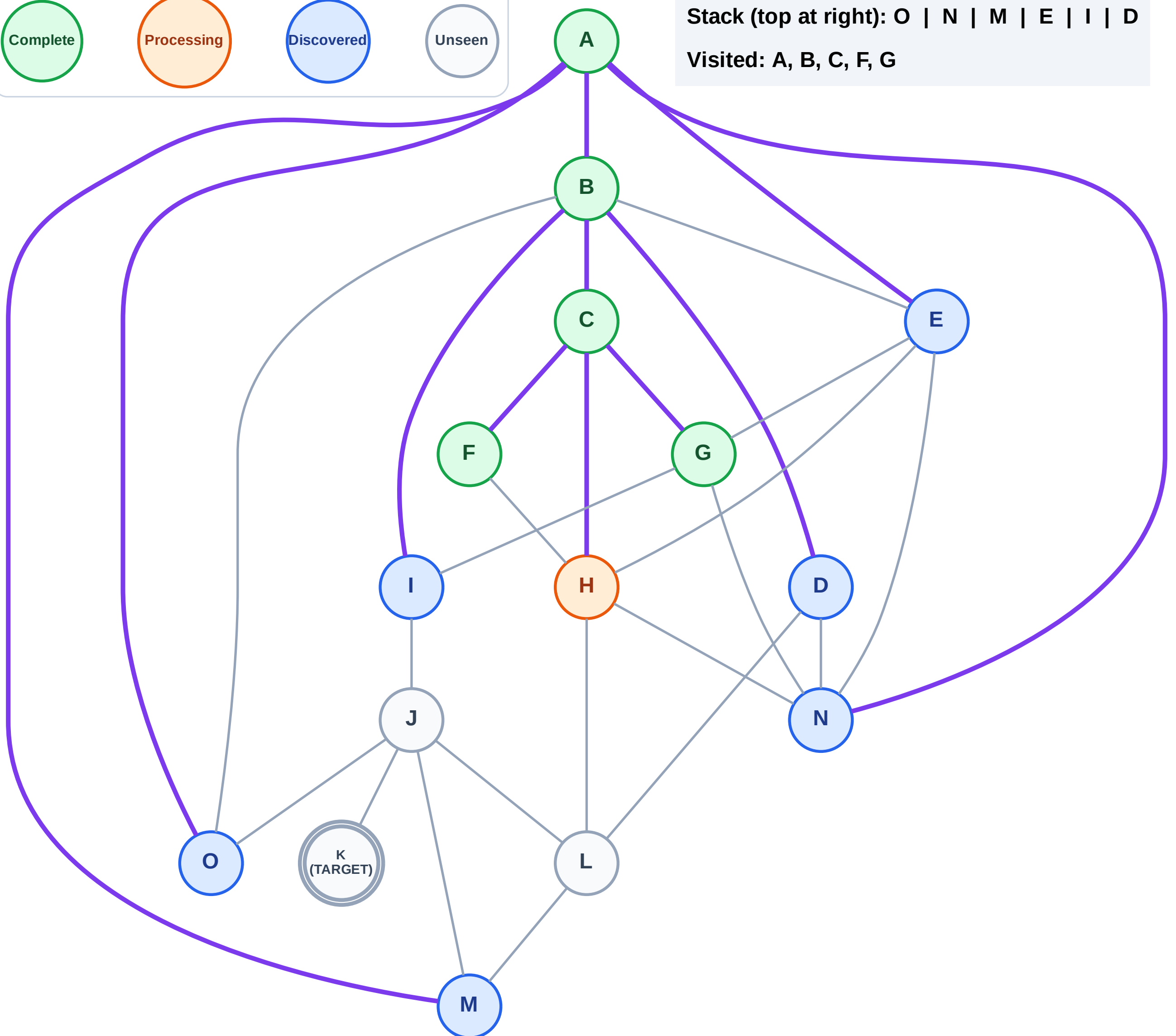
Step 12: Process node H

Legend



Stack (top at right): O | N | M | E | I | D

Visited: A, B, C, F, G



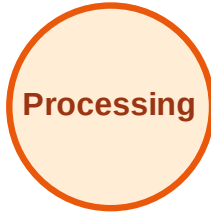
DFS Traversal

Step 13: Discover L from H

Legend



Complete



Processing



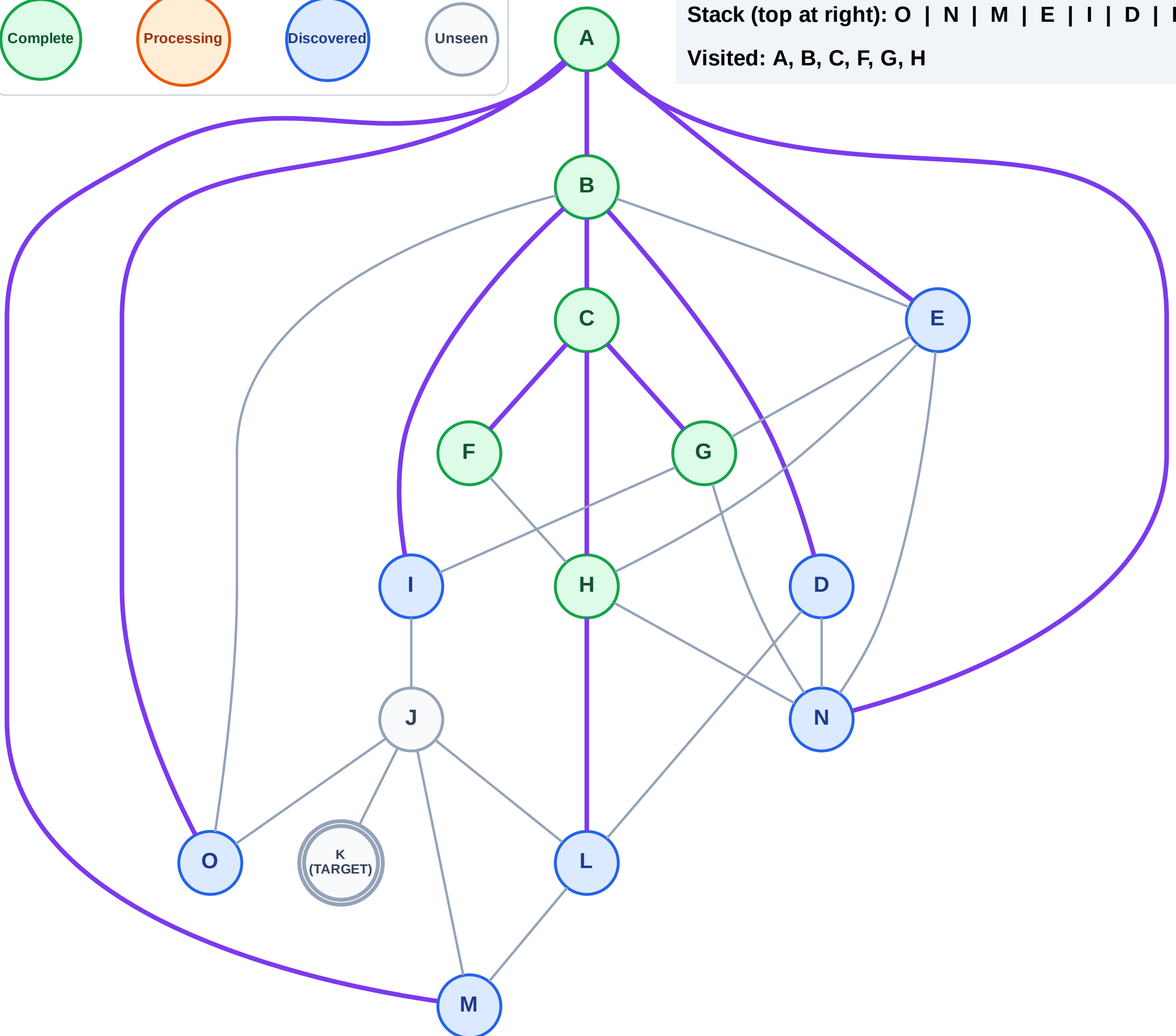
Discovered



Unseen

Stack (top at right): O | N | M | E | I | D | L

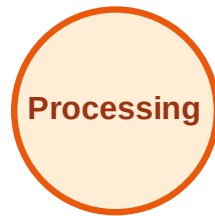
Visited: A, B, C, F, G, H



DFS Traversal

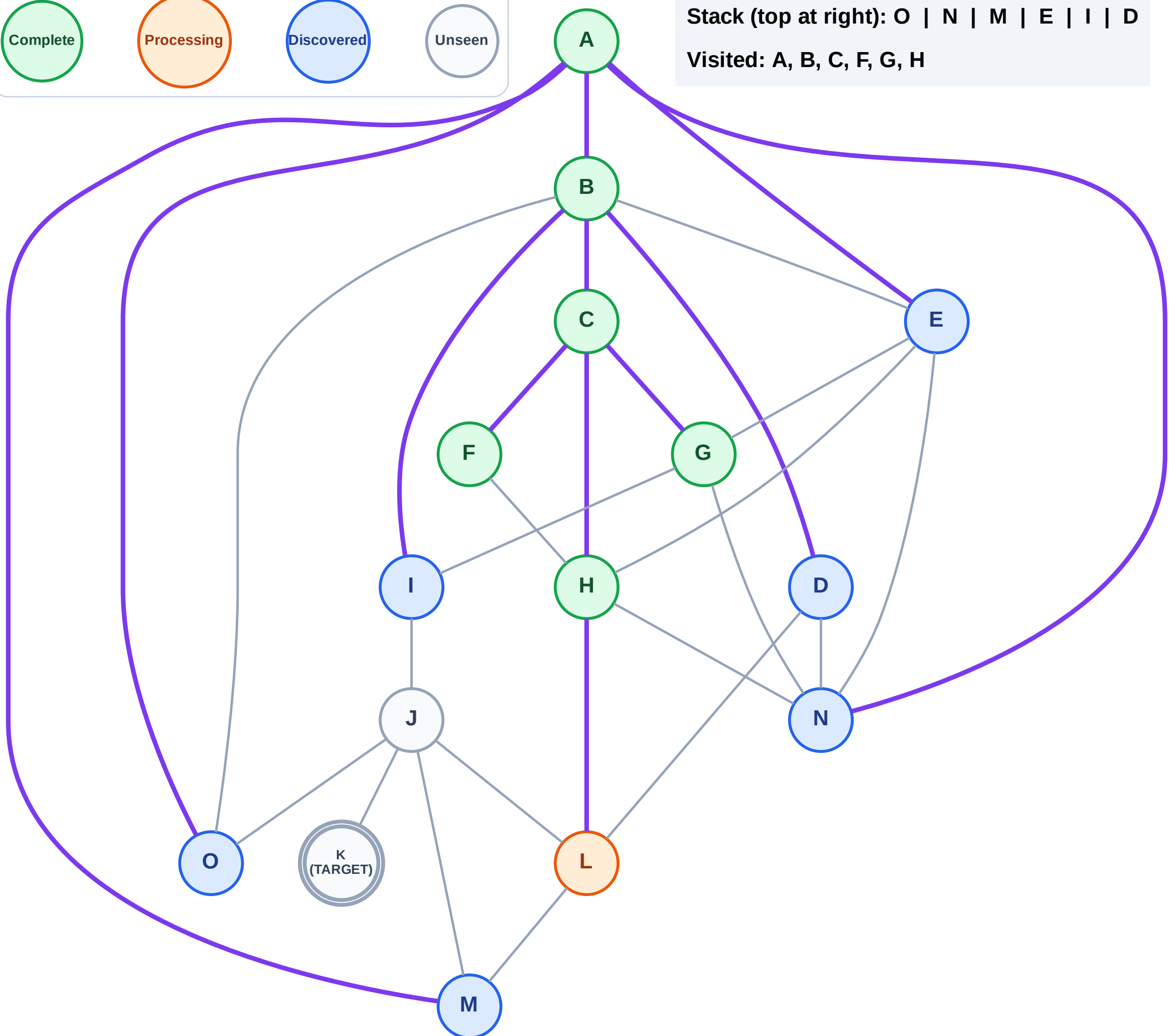
Step 14: Process node L

Legend



Stack (top at right): O | N | M | E | I | D

Visited: A, B, C, F, G, H



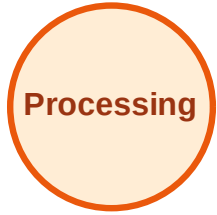
DFS Traversal

Step 15: Discover J from L

Legend



Complete



Processing



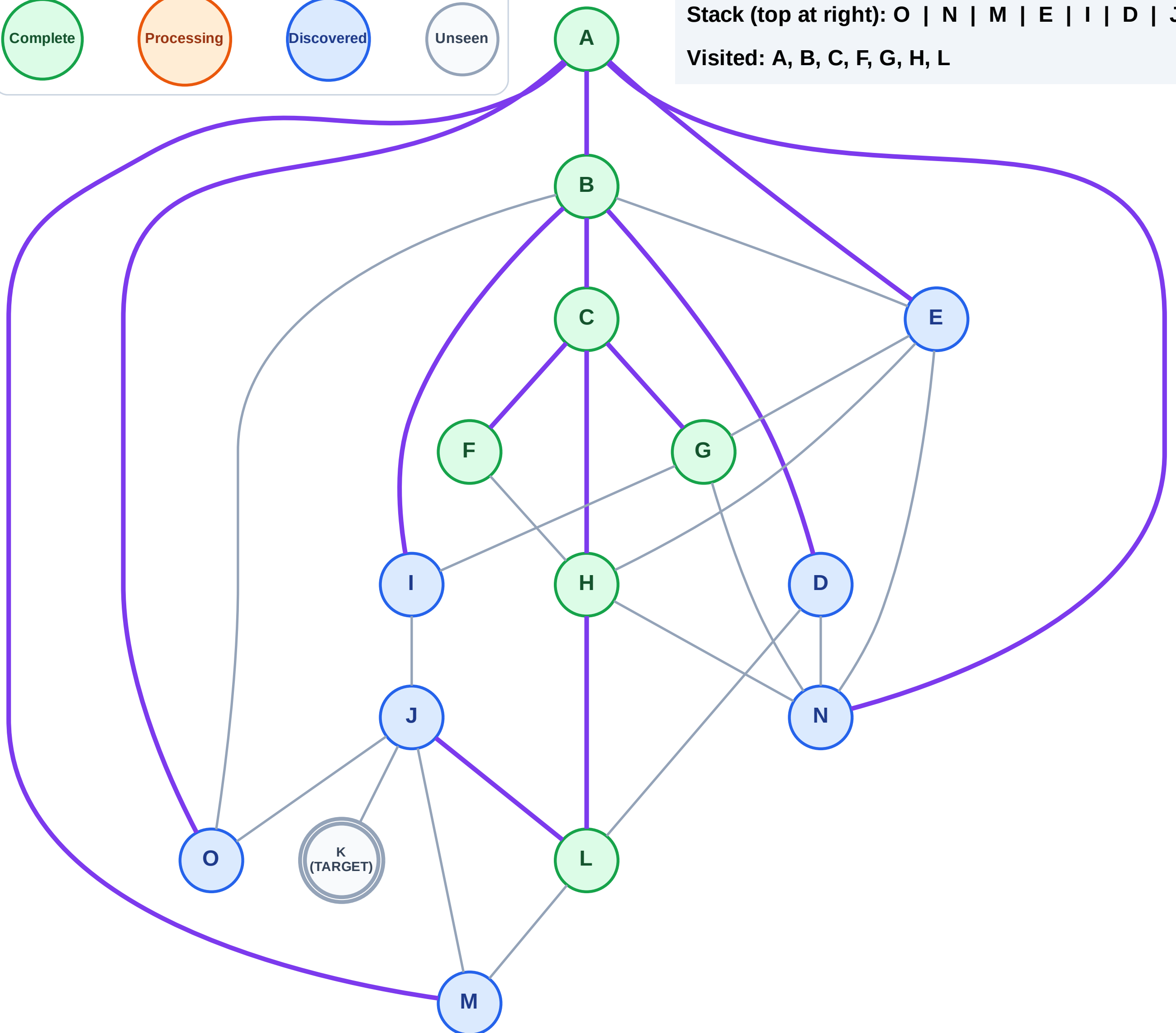
Discovered



Unseen

Stack (top at right): O | N | M | E | I | D | J

Visited: A, B, C, F, G, H, L



DFS Traversal

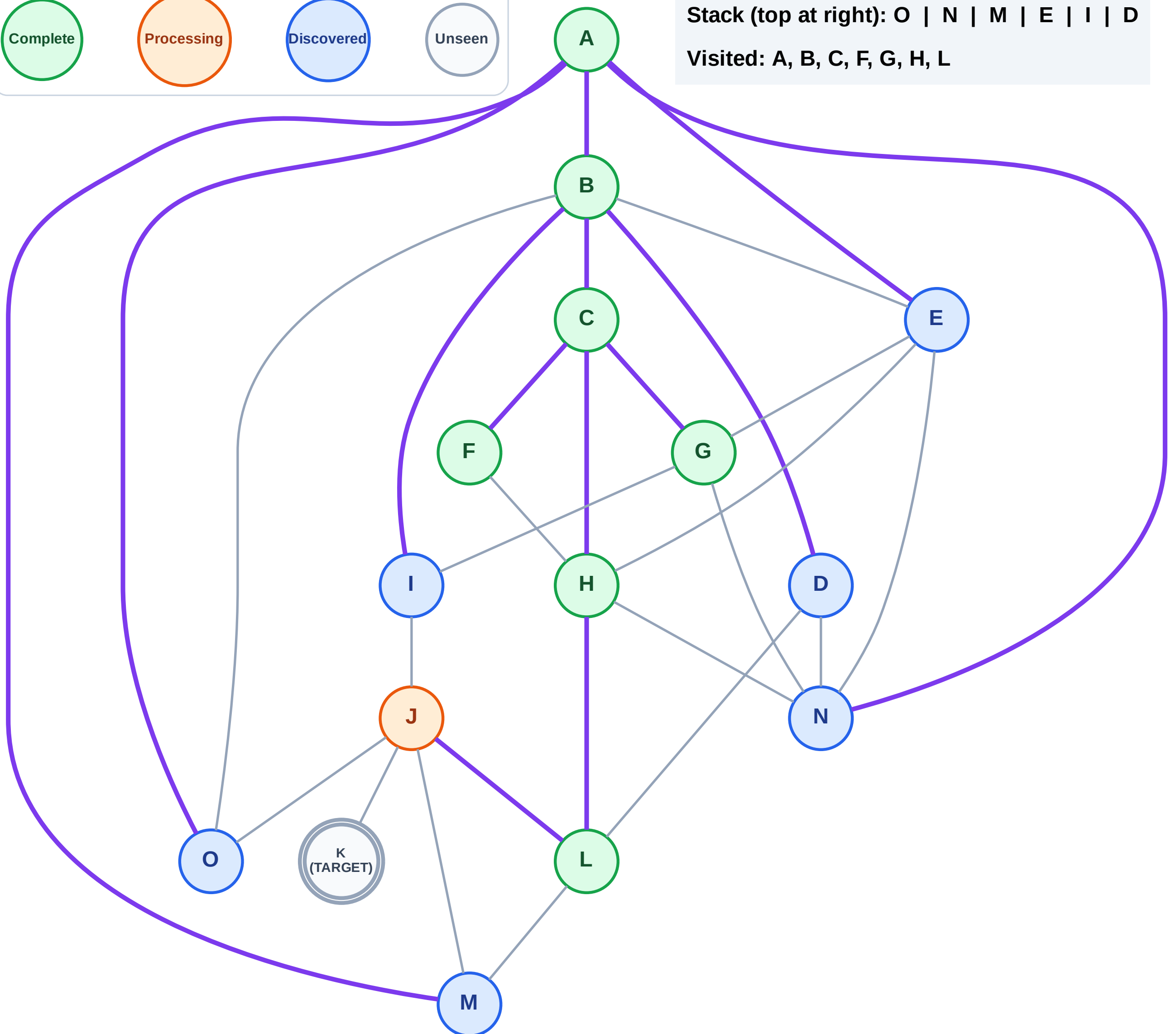
Step 16: Process node J

Legend



Stack (top at right): O | N | M | E | I | D

Visited: A, B, C, F, G, H, L



DFS Traversal

Step 17: TARGET NODE K FOUND (SHORT-CIRCUITED)! Path (6 hops): A → B → C → H → L → J → K

Legend

Complete

Processing

Discovered

Unseen

Stack (top at right): O | N | M | E | I | D
Visited: A, B, C, F, G, H, J, K, L

